

WE HAVE ALREADY INVESTIGATED SOME of the applications of derivatives, but now that we know the differentiation rules we are in a better position to pursue the applications of differentiation in greater depth. Here we learn how derivatives affect the shape of a graph of a function and, in particular, how they help us locate maximum and minimum values of functions. In addition, we use derivatives to provide insight into the long-term behavior of discrete-time models in the life sciences.

4.1 Maximum and Minimum Values

Some of the most important applications of differential calculus are *optimization problems*, in which we are required to find the optimal (best) way of doing something. Here are examples of such problems that we will solve in this chapter:

- What is the radius of a contracted windpipe that expels air most rapidly during a cough?
- At what angle should blood vessels branch so as to minimize the energy expended by the heart in pumping blood?

These problems can be reduced to finding the maximum or minimum values of a function. Let's first explain exactly what we mean by maximum and minimum values.

■ Absolute and Local Extreme Values

We see that the highest point on the graph of the function f shown in Figure 1 is the point $(3, 5)$. In other words, the largest value of f is $f(3) = 5$. Likewise, the smallest value is $f(6) = 2$. We say that $f(3) = 5$ is the *absolute maximum* of f and $f(6) = 2$ is the *absolute minimum*. In general, we use the following definition.

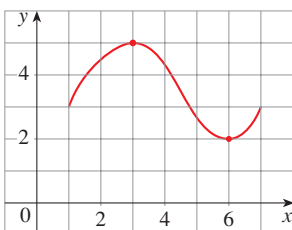


FIGURE 1

(1) Definition Let c be a number in the domain D of a function f . Then $f(c)$ is the

- **absolute maximum** value of f on D if $f(c) \geq f(x)$ for all x in D .
- **absolute minimum** value of f on D if $f(c) \leq f(x)$ for all x in D .

An absolute maximum or minimum is sometimes called a **global** maximum or minimum. The maximum and minimum values of f are called **extreme values** of f .

Figure 2 shows the graph of a function f with absolute maximum at d and absolute minimum at a . Note that $(d, f(d))$ is the highest point on the graph and $(a, f(a))$ is the lowest point. In Figure 2, if we consider only values of x near b [for instance, if we restrict our attention to the interval (a, c)], then $f(b)$ is the largest of those values of $f(x)$ and is called a *local maximum value* of f . Likewise, $f(c)$ is called a *local minimum value* of f because $f(c) \leq f(x)$ for x near c [in the interval (b, d) , for instance]. The function f also has a local minimum at e . In general, we have the following definition.

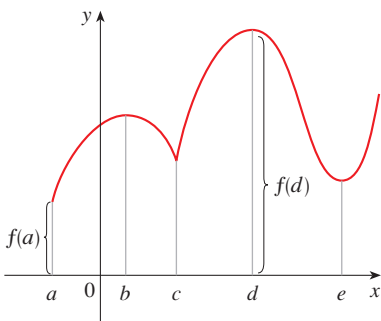


FIGURE 2

Abs min $f(a)$, abs max $f(d)$
loc min $f(c)$, $f(e)$, loc max $f(b)$, $f(d)$

(2) Definition The number $f(c)$ is a

- **local maximum** value of f if $f(c) \geq f(x)$ when x is near c .
- **local minimum** value of f if $f(c) \leq f(x)$ when x is near c .

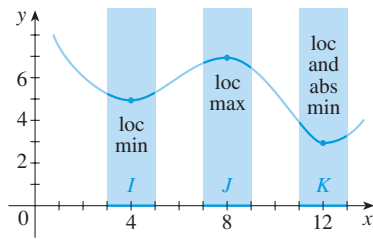
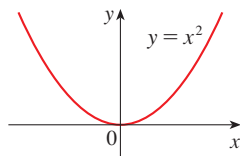


FIGURE 3

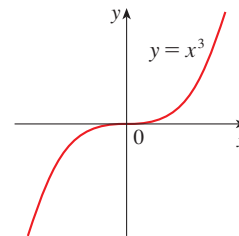
In Definition 2 (and elsewhere), if we say that something is true **near** c , we mean that it is true on some open interval containing c . For instance, in Figure 3 we see that $f(4) = 5$ is a local minimum because it's the smallest value of f on the interval I . It's not the absolute minimum because $f(x)$ takes smaller values when x is near 12 (in the interval K , for instance). In fact, $f(12) = 3$ is both a local minimum and the absolute minimum. Similarly, $f(8) = 7$ is a local maximum, but not the absolute maximum because f takes larger values near 1.

EXAMPLE 1 | The function $f(x) = \cos x$ takes on its (local and absolute) maximum value of 1 infinitely many times, since $\cos 2n\pi = 1$ for any integer n and $-1 \leq \cos x \leq 1$ for all x . Likewise, $\cos(2n + 1)\pi = -1$ is its minimum value, where n is any integer.


 FIGURE 4
Minimum value 0, no maximum

EXAMPLE 2 | If $f(x) = x^2$, then $f(x) \geq f(0)$ because $x^2 \geq 0$ for all x . Therefore $f(0) = 0$ is the absolute (and local) minimum value of f . This corresponds to the fact that the origin is the lowest point on the parabola $y = x^2$. (See Figure 4.) However, there is no highest point on the parabola and so this function has no maximum value.

EXAMPLE 3 | From the graph of the function $f(x) = x^3$, shown in Figure 5, we see that this function has neither an absolute maximum value nor an absolute minimum value. In fact, it has no local extreme values either.


 FIGURE 5
No minimum, no maximum

EXAMPLE 4 | **BB** **Electrocardiogram** Figure 6 shows a rhythm strip from an ECG. It is a graph of the electric potential function $f(t)$ (measured in millivolts) as a function of time in a certain direction corresponding to a particular part of the heart. The points P , Q , R , S , and T on the graph are labeled with the notation that is standard for cardiologists. We see that the function f has local maxima at the points P , R , and T , with an absolute maximum at R , and local minima at Q and S , with an absolute minimum at S .

Cardiologists use the relative locations of the extreme points P , Q , R , S , and T to diagnose heart problems.

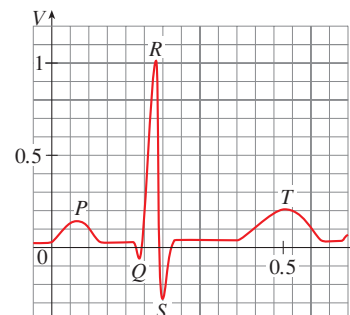


FIGURE 6

Each square in the grid corresponds to a horizontal distance of 0.04 seconds and a vertical distance of 0.1 mV. So the maximum and minimum values in this particular ECG are as follows.

Local maximum values: At P , $f(0.05) \approx 0.15$ mV; at R , $f(0.22) \approx 1.0$ mV;
at T , $f(0.5) \approx 0.2$ mV

Absolute maximum value: At R , $f(0.22) \approx 1.0$ mV

Local minimum values: At Q , $f(0.19) \approx -0.06$ mV; at S , $f(0.24) \approx -0.3$ mV

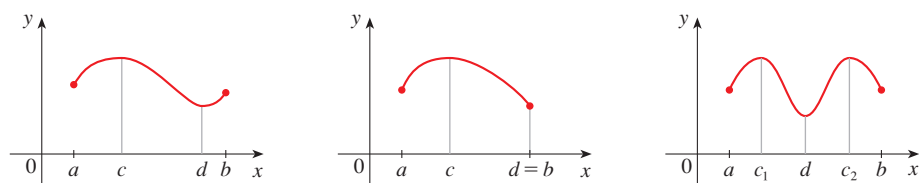
Absolute minimum value: At S , $f(0.24) \approx -0.3$ mV

We have seen that some functions have extreme values, whereas others do not. The following theorem gives conditions under which a function is guaranteed to possess extreme values.

(3) The Extreme Value Theorem If f is continuous on a closed interval $[a, b]$, then f attains an absolute maximum value $f(c)$ and an absolute minimum value $f(d)$ at some numbers c and d in $[a, b]$.

The Extreme Value Theorem is illustrated in Figure 7. Note that an extreme value can be taken on more than once. Although the Extreme Value Theorem is intuitively very plausible, it is difficult to prove and so we omit the proof.

FIGURE 7



Figures 8 and 9 show that a function need not possess extreme values if either hypothesis (continuity or closed interval) is omitted from the Extreme Value Theorem.

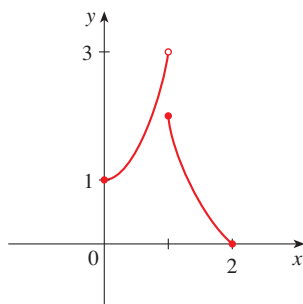


FIGURE 8

This function has a minimum value $f(2) = 0$, but no maximum value.

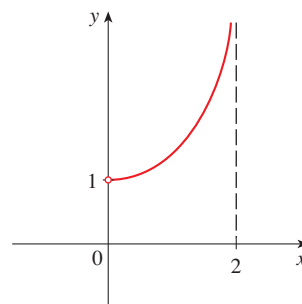


FIGURE 9

This continuous function g has no maximum or minimum.

The function f whose graph is shown in Figure 8 is defined on the closed interval $[0, 2]$ but has no maximum value. [Notice that the range of f is $[0, 3)$. The function takes on values arbitrarily close to 3, but never actually attains the value 3.] This does

not contradict the Extreme Value Theorem because f is not continuous. [Nonetheless, a discontinuous function *could* have maximum and minimum values. See Exercise 13(c).]

The function g shown in Figure 9 is continuous on the open interval $(0, 2)$ but has neither a maximum nor a minimum value. [The range of g is $(1, \infty)$. The function takes on arbitrarily large values.] This does not contradict the Extreme Value Theorem because the interval $(0, 2)$ is not closed.

Fermat's Theorem

The Extreme Value Theorem says that a continuous function on a closed interval has a maximum value and a minimum value, but it does not tell us how to find these extreme values. We start by looking for local extreme values.

Figure 10 shows the graph of a function f with a local maximum at c and a local minimum at d . It appears that at the maximum and minimum points the tangent lines are horizontal and therefore each has slope 0. We know that the derivative is the slope of the tangent line, so it appears that $f'(c) = 0$ and $f'(d) = 0$. The following theorem says that this is always true for differentiable functions.

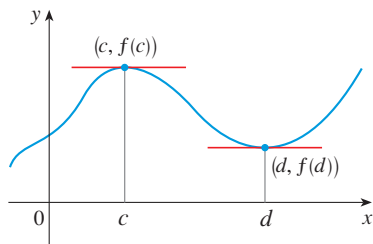


FIGURE 10

(4) Fermat's Theorem If f has a local maximum or minimum at c , and if $f'(c)$ exists, then $f'(c) = 0$.

Our intuition suggests that Fermat's Theorem is true. A rigorous proof, using the definition of a derivative, is given in Appendix E.

Although Fermat's Theorem is very useful, we have to guard against reading too much into it. If $f(x) = x^3$, then $f'(x) = 3x^2$, so $f'(0) = 0$. But f has no maximum or minimum at 0, as you can see from its graph in Figure 11. The fact that $f'(0) = 0$ simply means that the curve $y = x^3$ has a horizontal tangent at $(0, 0)$. Instead of having a maximum or minimum at $(0, 0)$, the curve crosses its horizontal tangent there.



Thus, when $f'(c) = 0$, f doesn't necessarily have a maximum or minimum at c . (In other words, the converse of Fermat's Theorem is false in general.)

Fermat

Fermat's Theorem is named after Pierre Fermat (1601–1665), a French lawyer who took up mathematics as a hobby. Despite his amateur status, Fermat was one of the two inventors of analytic geometry (Descartes was the other). His methods for finding tangents to curves and maximum and minimum values (before the invention of limits and derivatives) made him a forerunner of Newton in the creation of differential calculus.

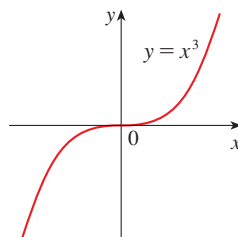


FIGURE 11

If $f(x) = x^3$, then $f'(0) = 0$ but f has no maximum or minimum.

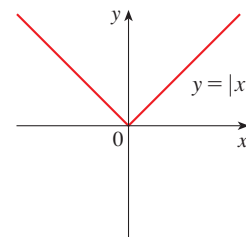


FIGURE 12

If $f(x) = |x|$, then $f(0) = 0$ is a minimum value, but $f'(0)$ does not exist.

We should bear in mind that there may be an extreme value where $f'(c)$ does not exist. For instance, the function $f(x) = |x|$ has its (local and absolute) minimum value at 0 (see Figure 12), but that value cannot be found by setting $f'(x) = 0$ because, as was shown in Example 3.2.6, $f'(0)$ does not exist.

Fermat's Theorem does suggest that we should at least *start* looking for extreme values of f at the numbers c where $f'(c) = 0$ or where $f'(c)$ does not exist. Such numbers are given a special name.

(5) Definition A **critical number** of a function f is a number c in the domain of f such that either $f'(c) = 0$ or $f'(c)$ does not exist.

Figure 13 shows a graph of the function f in Example 5. It supports our answer because there is a horizontal tangent when $x = 1.5$ and a vertical tangent when $x = 0$.

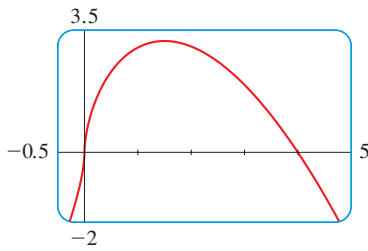


FIGURE 13

EXAMPLE 5 | Find the critical numbers of $f(x) = x^{3/5}(4 - x)$.

SOLUTION The Product Rule gives

$$\begin{aligned} f'(x) &= x^{3/5}(-1) + (4 - x)\left(\frac{3}{5}x^{-2/5}\right) = -x^{3/5} + \frac{3(4 - x)}{5x^{2/5}} \\ &= \frac{-5x + 3(4 - x)}{5x^{2/5}} = \frac{12 - 8x}{5x^{2/5}} \end{aligned}$$

[The same result could be obtained by first writing $f(x) = 4x^{3/5} - x^{8/5}$.] Therefore $f'(x) = 0$ if $12 - 8x = 0$, that is, $x = \frac{3}{2}$, and $f'(x)$ does not exist when $x = 0$. Thus the critical numbers are $\frac{3}{2}$ and 0. ■

In terms of critical numbers, Fermat's Theorem can be rephrased as follows (compare Definition 5 with Theorem 4):

(6) If f has a local maximum or minimum at c , then c is a critical number of f .

■ The Closed Interval Method

To find an absolute maximum or minimum of a continuous function on a closed interval, we note that either it is local [in which case it occurs at a critical number by (6)] or it occurs at an endpoint of the interval. Thus the following three-step procedure always works.

The Closed Interval Method To find the *absolute* maximum and minimum values of a continuous function f on a closed interval $[a, b]$:

1. Find the values of f at the critical numbers of f in (a, b) .
2. Find the values of f at the endpoints of the interval.
3. The largest of the values from Steps 1 and 2 is the absolute maximum value; the smallest of these values is the absolute minimum value.

The phenomenon whereby a population declines to extinction below a critical population size is referred to as an *Allee effect* after the American ecologist Warder Clyde Allee (1885–1955).

EXAMPLE 6 | **BB** **The Allee effect** One of the models for the growth rate of a population of size N at time t reflects the fact that some populations decline to extinction unless they stay above a critical value. A particular case of this model is expressed by the growth rate

$$f(N) = N(N - 3)(8 - N)$$

where N is measured in hundreds of individuals. [Notice that $f(N)$ is negative when $0 < N < 3$.] Find the absolute maximum and minimum values of the growth rate function

$$f(N) = N(N - 3)(8 - N) \quad 0 \leq N \leq 9$$

SOLUTION Because f is continuous on the interval $[0, 9]$, we can use the Closed Interval Method:

$$f(N) = N(N - 3)(8 - N) = -N^3 + 11N^2 - 24N$$

$$f'(N) = -3N^2 + 22N - 24 = -(3N - 4)(N - 6)$$

Since $f'(N)$ exists for all N , the only critical numbers of f occur when $f'(N) = 0$, that is, $N = \frac{4}{3}$ or $N = 6$. Notice that each of these critical numbers lies in the interval $(0, 9)$. The values of f at these critical numbers are

$$f\left(\frac{4}{3}\right) = -\frac{400}{27} \quad f(6) = 36$$

The values of f at the endpoints of the interval are

$$f(0) = 0 \quad f(9) = -54$$

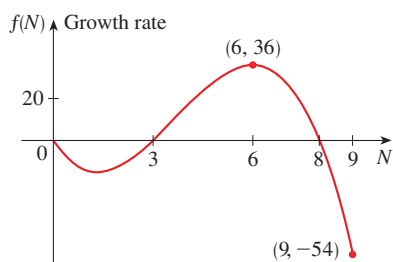


FIGURE 14

Comparing these four numbers, we see that the absolute maximum value is $f(6) = 36$ and the absolute minimum value is $f(9) = -54$.

So the population increases fastest when $N = 6$ (the population is 600) and the absolute maximum value is $f(6) = 36$, which means that the maximum rate of increase is 3600 individuals per year. The population decreases most rapidly on the given interval when $N = 9$ and the absolute minimum value is $f(9) = -54$. This means that the maximum rate of decrease is 5400 individuals per year.

Note that in this example the absolute minimum occurs at an endpoint, whereas the absolute maximum occurs at a critical number. The graph of f is sketched in Figure 14.

EXAMPLE 7 | Blood alcohol concentration In Section 3.1 we used the function

$$C(t) = 0.0225te^{-0.0467t}$$

to model the average blood alcohol concentration (BAC) of a group of eight male subjects after rapid consumption of 15 mL of ethanol (corresponding to one alcoholic drink), where t is measured in minutes after consumption and C is measured in mg/mL. Find the maximum value of the BAC during the first hour.

SOLUTION We begin by differentiating the concentration function using the Product Rule:

$$\begin{aligned} C'(t) &= 0.0225t(-0.0467)e^{-0.0467t} + 0.0225e^{-0.0467t} \\ &= 0.0225e^{-0.0467t}(-0.0467t + 1) \end{aligned}$$

The critical number occurs when $C'(t) = 0$, that is,

$$0.0467t = 1 \quad \Rightarrow \quad t = \frac{1}{0.0467} \approx 21.4$$

The value of C at this critical number is about

$$C(21.4) \approx 0.177$$

and the values of C at the endpoints of the interval $[0, 60]$ are

$$C(0) = 0 \quad C(60) \approx 0.0819$$

Comparing the values of C at the critical number and at the endpoints, we see that the maximum value of the BAC in the first hour was about 0.177 mg/mL and this occurred about 21 minutes after consumption. (See the graph of C in Figure 15.) Notice that the maximum value of 0.177 mg/mL was well above the legal driving limit of 0.08 mg/mL and occurred after just one drink.

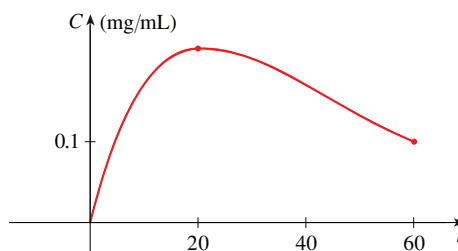
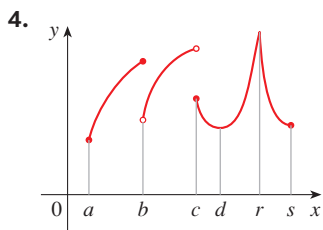
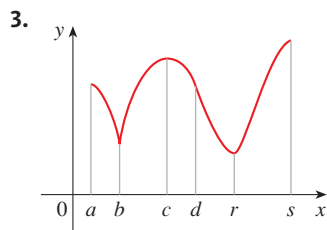


FIGURE 15

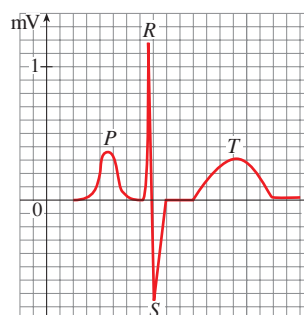
EXERCISES 4.1

1. Explain the difference between an absolute minimum and a local minimum.
2. Suppose f is a continuous function defined on a closed interval $[a, b]$.
 - (a) What theorem guarantees the existence of an absolute maximum value and an absolute minimum value for f ?
 - (b) What steps would you take to find those maximum and minimum values?

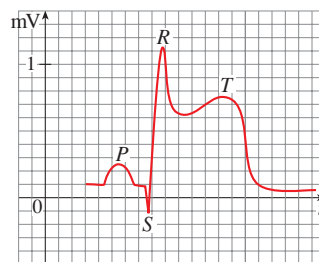
3–4 For each of the numbers a, b, c, d, r , and s , state whether the function whose graph is shown has an absolute maximum or minimum, a local maximum or minimum, or neither a maximum nor a minimum.



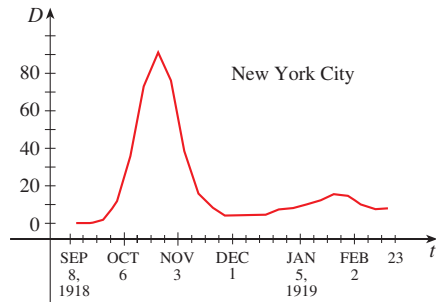
5. **Electrocardiogram** A cardiologist looking at the rhythm strip shown might suspect *right atrial hypertrophy* because of the relatively tall peaked wave at P (compare with Figure 6). State the local and absolute maximum and minimum values of the electric potential function $f(t)$.

1 square = 0.04 s $\times$ 0.1 mV

6. **Electrocardiogram** A cardiologist looking at this rhythm strip might suspect *infarction* because of the elevation of the graph near S and T (compare with Figure 6). State the local and absolute maximum and minimum values of the electric potential function $f(t)$.

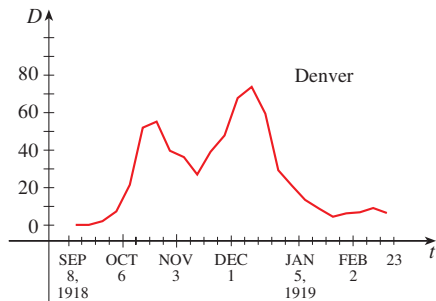
1 square = 0.04 s $\times$ 0.1 mV

7. In the **influenza pandemic** of 1918–1919 about 40 million people died worldwide. A study in 2007 assessed the nonpharmaceutical interventions used in 43 US cities to combat the infection, including isolation, quarantines, school closures, and public gathering cancellations. The graph shows the weekly excess death rate $D(t)$ for New York City. State the local and absolute maximum and minimum values of D over the given time period and estimate when they occurred.



Source: H. Markel et al., “Nonpharmaceutical Interventions Implemented by US Cities During the 1918–1919 Influenza Pandemic,” *J. Amer. Med. Assn.* 298 (2007): 644–54.

8. **Influenza pandemic** The study cited in Exercise 7 also included the corresponding graph for Denver shown here.
- State the corresponding local and absolute maximum and minimum values for Denver.
 - Compare the graphs for New York and Denver. How do you think the strategies differed in the two cities?



9–12 Sketch the graph of a function f that is continuous on $[1, 5]$ and has the given properties.

- Absolute minimum at 2, absolute maximum at 3, local minimum at 4
- Absolute minimum at 1, absolute maximum at 5, local maximum at 2, local minimum at 4
- Absolute maximum at 5, absolute minimum at 2, local maximum at 3, local minima at 2 and 4
- f has no local maximum or minimum, but 2 and 4 are critical numbers

- Sketch the graph of a function that has a local maximum at 2 and is differentiable at 2.
 - Sketch the graph of a function that has a local maximum at 2 and is continuous but not differentiable at 2.
 - Sketch the graph of a function that has a local maximum at 2 and is not continuous at 2.
- Sketch the graph of a function on $[-1, 2]$ that has an absolute maximum but no local maximum.
 - Sketch the graph of a function on $[-1, 2]$ that has a local maximum but no absolute maximum.
- Sketch the graph of a function on $[-1, 2]$ that has an absolute maximum but no absolute minimum.
 - Sketch the graph of a function on $[-1, 2]$ that is discontinuous but has both an absolute maximum and an absolute minimum.
- Sketch the graph of a function that has two local maxima, one local minimum, and no absolute minimum.
 - Sketch the graph of a function that has three local minima, two local maxima, and seven critical numbers.

17–24 Sketch the graph of f by hand and use your sketch to find the absolute and local maximum and minimum values of f . (Use the graphs and transformations of Sections 1.2 and 1.3.)

- $f(x) = \frac{1}{2}(3x - 1), \quad x \leq 3$
- $f(x) = 2 - \frac{1}{3}x, \quad x \geq -2$
- $f(x) = x^2, \quad 0 < x < 2$
- $f(x) = e^x$
- $f(x) = \ln x, \quad 0 < x \leq 2$
- $f(t) = \cos t, \quad -3\pi/2 \leq t \leq 3\pi/2$
- $f(x) = 1 - \sqrt{x}$
- $f(x) = \begin{cases} 4 - x^2 & \text{if } -2 \leq x < 0 \\ 2x - 1 & \text{if } 0 \leq x \leq 2 \end{cases}$

25–40 Find the critical numbers of the function.

- $f(x) = 4 + \frac{1}{3}x - \frac{1}{2}x^2$
- $f(x) = x^3 + 6x^2 - 15x$
- $f(x) = x^3 + 3x^2 - 24x$
- $f(x) = x^3 + x^2 + x$
- $s(t) = 3t^4 + 4t^3 - 6t^2$
- $g(t) = |3t - 4|$
- $g(y) = \frac{y - 1}{y^2 - y + 1}$
- $h(p) = \frac{p - 1}{p^2 + 4}$
- $h(t) = t^{3/4} - 2t^{1/4}$
- $g(x) = x^{1/3} - x^{-2/3}$
- $F(x) = x^{4/5}(x - 4)^2$
- $g(\theta) = 4\theta - \tan \theta$
- $f(\theta) = 2 \cos \theta + \sin^2 \theta$
- $h(t) = 3t - \arcsin t$
- $f(x) = x^2 e^{-3x}$
- $f(x) = x^{-2} \ln x$

41–54 Find the absolute maximum and absolute minimum values of f on the given interval.

41. $f(x) = 12 + 4x - x^2$, $[0, 5]$

42. $f(x) = 5 + 54x - 2x^3$, $[0, 4]$

43. $f(x) = 2x^3 - 3x^2 - 12x + 1$, $[-2, 3]$

44. $f(x) = x^3 - 6x^2 + 9x + 2$, $[-1, 4]$

45. $f(x) = x^4 - 2x^2 + 3$, $[-2, 3]$

46. $f(x) = (x^2 - 1)^3$, $[-1, 2]$

47. $f(t) = t\sqrt{4 - t^2}$, $[-1, 2]$

48. $f(x) = \frac{x^2 - 4}{x^2 + 4}$, $[-4, 4]$

49. $f(x) = xe^{-x^2/8}$, $[-1, 4]$

50. $f(x) = x - \ln x$, $[\frac{1}{2}, 2]$

51. $f(x) = \ln(x^2 + x + 1)$, $[-1, 1]$

52. $f(x) = x - 2 \tan^{-1}x$, $[0, 4]$

53. $f(t) = 2 \cos t + \sin 2t$, $[0, \pi/2]$

54. $f(t) = t + \cot(t/2)$, $[\pi/4, 7\pi/4]$

55. If a and b are positive numbers, find the maximum value of $f(x) = x^a(1 - x)^b$, $0 \leq x \leq 1$.

56. Antibiotic pharmacokinetics After an antibiotic tablet is taken, the concentration of the antibiotic in the bloodstream is modeled by the function

$$C(t) = 8(e^{-0.4t} - e^{-0.6t})$$

where the time t is measured in hours and C is measured in $\mu\text{g/mL}$. What is the maximum concentration of the antibiotic during the first 12 hours?

57. Disease virulence The Kermack-McKendrick model for infectious disease transmission (see Exercise 7.6.23) can be used to predict the population size P as a function of the disease's *virulence* (that is, the extent to which the disease kills people). The population size P is large when virulence v is low and it is also large when virulence is high because the disease kills people so fast that very few people get infected. For a specific choice of constants, the population size is

$$P(v) = \frac{10 + v + v^2}{1 + v} \quad 0 \leq v \leq 9$$

Find the smallest and largest population sizes and the virulence values for which they occur.

58. The Maynard Smith and Slatkin model for population growth is a discrete-time model of the form

$$n_{t+1} = \frac{\lambda n_t}{1 + \alpha n_t^k}$$

For the constants $\lambda = 2$, $\alpha = 0.25$, and $k = 2$, the model is

$n_{t+1} = f(n_t)$, where the updating function is

$$f(n) = \frac{2n}{1 + 0.25n^2}$$

Find the largest value of f and interpret it. [Hint: Consider $\lim_{n \rightarrow \infty} f(n)$.]

59. Coughing When a foreign object lodged in the trachea (windpipe) forces a person to cough, the diaphragm thrusts upward causing an increase in pressure in the lungs. This is accompanied by a contraction of the trachea, making a narrower channel for the expelled air to flow through. For a given amount of air to escape in a fixed time, it must move faster through the narrower channel than the wider one. The greater the velocity of the airstream, the greater the force on the foreign object. X rays show that the radius of the circular tracheal tube contracts to about two-thirds of its normal radius during a cough. According to a mathematical model of coughing, the velocity v of the airstream is related to the radius r of the trachea by the equation

$$v(r) = k(r_0 - r)r^2 \quad \frac{1}{2}r_0 \leq r \leq r_0$$

where k is a constant and r_0 is the normal radius of the trachea. The restriction on r is due to the fact that the tracheal wall stiffens under pressure and a contraction greater than $\frac{1}{2}r_0$ is prevented (otherwise the person would suffocate).

- Determine the value of r in the interval $[\frac{1}{2}r_0, r_0]$ at which v has an absolute maximum. How does this compare with experimental evidence?
- What is the absolute maximum value of v on the interval?
- Sketch the graph of v on the interval $[0, r_0]$.

 **60.** On May 7, 1992, the space shuttle *Endeavour* was launched on mission STS-49, the purpose of which was to install a new perigee kick motor in an Intelsat communications satellite. The table gives the velocity data for the shuttle between liftoff and the jettisoning of the solid rocket boosters.

Event	Time (s)	Velocity (ft/s)
Launch	0	0
Begin roll maneuver	10	185
End roll maneuver	15	319
Throttle to 89%	20	447
Throttle to 67%	32	742
Throttle to 104%	59	1325
Maximum dynamic pressure	62	1445
Solid rocket booster separation	125	4151

- Use a graphing calculator or computer to find the cubic polynomial that best models the velocity of the shuttle for the time interval $t \in [0, 125]$. Then graph this polynomial.
- Find a model for the acceleration of the shuttle and use it to estimate the maximum and minimum values of the acceleration during the first 125 seconds.

61. Between 0°C and 30°C the volume V (in cubic centimeters) of 1 kg of water at a temperature T is given approximately by the formula

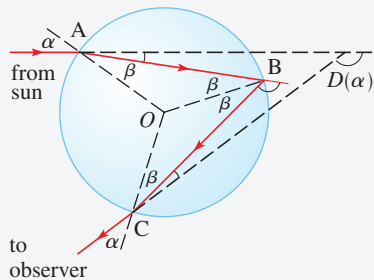
$$V = 999.87 - 0.06426T + 0.0085043T^2 - 0.0000679T^3$$

Find the temperature at which water has its maximum density.

62. A cubic function is a polynomial of degree 3; that is, it has the form $f(x) = ax^3 + bx^2 + cx + d$, where $a \neq 0$.
- Show that a cubic function can have two, one, or no critical number(s). Give examples and sketches to illustrate the three possibilities.
 - How many local extreme values can a cubic function have?

PROJECT The Calculus of Rainbows

Rainbows are created when raindrops scatter sunlight. They have fascinated humankind since ancient times and have inspired attempts at scientific explanation since the time of Aristotle. In this project we use the ideas of Descartes and Newton to explain the shape, location, and colors of rainbows.



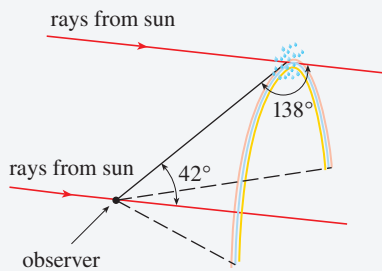
Formation of the primary rainbow

- The figure shows a ray of sunlight entering a spherical raindrop at A. Some of the light is reflected, but the line AB shows the path of the part that enters the drop. Notice that the light is refracted toward the normal line AO and in fact Snell's Law says that $\sin \alpha = k \sin \beta$, where α is the angle of incidence, β is the angle of refraction, and $k \approx \frac{4}{3}$ is the index of refraction for water. At B some of the light passes through the drop and is refracted into the air, but the line BC shows the part that is reflected. (The angle of incidence equals the angle of reflection.) When the ray reaches C, part of it is reflected, but for the time being we are more interested in the part that leaves the raindrop at C. (Notice that it is refracted away from the normal line.) The *angle of deviation* $D(\alpha)$ is the amount of clockwise rotation that the ray has undergone during this three-stage process. Thus

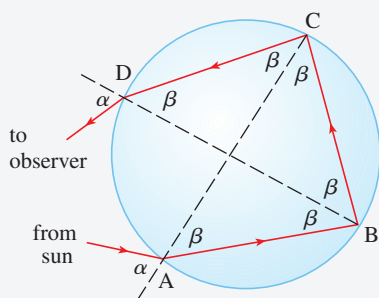
$$D(\alpha) = (\alpha - \beta) + (\pi - 2\beta) + (\alpha - \beta) = \pi + 2\alpha - 4\beta$$

Show that the minimum value of the deviation is $D(\alpha) \approx 138^\circ$ and occurs when $\alpha \approx 59.4^\circ$.

The significance of the minimum deviation is that when $\alpha \approx 59.4^\circ$ we have $D'(\alpha) \approx 0$, so $\Delta D / \Delta \alpha \approx 0$. This means that many rays with $\alpha \approx 59.4^\circ$ become deviated by approximately the same amount. It is the *concentration* of rays coming from near the direction of minimum deviation that creates the brightness of the primary rainbow. The figure at the left shows that the angle of elevation from the observer up to the highest point on the rainbow is $180^\circ - 138^\circ = 42^\circ$. (This angle is called the *rainbow angle*.)



- Problem 1 explains the location of the primary rainbow, but how do we explain the colors? Sunlight comprises a range of wavelengths, from the red range through orange, yellow, green, blue, indigo, and violet. As Newton discovered in his prism experiments of 1666, the index of refraction is different for each color. (The effect is called *dispersion*.) For red light the refractive index is $k \approx 1.3318$ whereas for violet light it is $k \approx 1.3435$. By repeating the calculation of Problem 1 for these values of k , show that the rainbow angle is about 42.3° for the red bow and 40.6° for the violet bow. So the rainbow really consists of seven individual bows corresponding to the seven colors.



Formation of the secondary rainbow

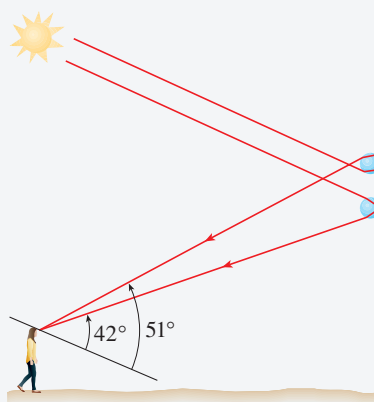
3. Perhaps you have seen a fainter secondary rainbow above the primary bow. That results from the part of a ray that enters a raindrop and is refracted at A, reflected twice (at B and C), and refracted as it leaves the drop at D (see the figure at the left). This time the deviation angle $D(\alpha)$ is the total amount of counterclockwise rotation that the ray undergoes in this four-stage process. Show that

$$D(\alpha) = 2\alpha - 6\beta + 2\pi$$

and $D(\alpha)$ has a minimum value when

$$\cos \alpha = \sqrt{\frac{k^2 - 1}{8}}$$

Taking $k = \frac{4}{3}$, show that the minimum deviation is about 129° and so the rainbow angle for the secondary rainbow is about 51° , as shown in the following figure.



4. Show that the colors in the secondary rainbow appear in the opposite order from those in the primary rainbow.



4.2 How Derivatives Affect the Shape of a Graph

Many of the applications of calculus depend on our ability to deduce facts about a function f from information concerning its derivatives. At the end of Section 3.2 we discussed one instance of this principle by conjecturing that if f has a positive derivative, then it is an increasing function. Here we prove that fact and also see how the second derivative of a function influences the shape of its graph.

Lagrange and the Mean Value Theorem

The Mean Value Theorem was first formulated by Joseph-Louis Lagrange (1736–1813), born in Italy of a French father and an Italian mother. He was a child prodigy and became a professor in Turin at the tender age of 19. Lagrange made great contributions to number theory, theory of functions, theory of equations, and analytical and celestial mechanics. In particular, he applied calculus to the analysis of the stability of the solar system. At the invitation of Frederick the Great, he succeeded Euler at the Berlin Academy and, when Frederick died, Lagrange accepted King Louis XVI's invitation to Paris, where he was given apartments in the Louvre and became a professor at the Ecole Polytechnique. Despite all the trappings of luxury and fame, he was a kind and quiet man, living only for science.

The Mean Value Theorem

We start with a fact, known as the Mean Value Theorem, that will be useful not only for present purposes but also for explaining why some of the other basic results of calculus are true.

The Mean Value Theorem If f is a differentiable function on the interval $[a, b]$, then there exists a number c between a and b such that

$$(1) \quad f'(c) = \frac{f(b) - f(a)}{b - a}$$

or, equivalently,

$$(2) \quad f(b) - f(a) = f'(c)(b - a)$$

We can see that this theorem is reasonable by interpreting it geometrically. Figures 1 and 2 show the points $A(a, f(a))$ and $B(b, f(b))$ on the graphs of two differentiable functions.

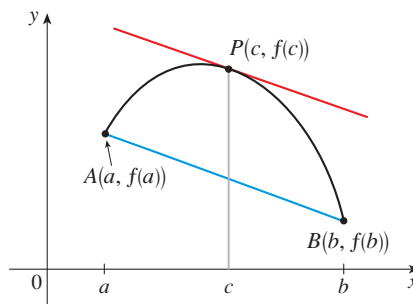


FIGURE 1

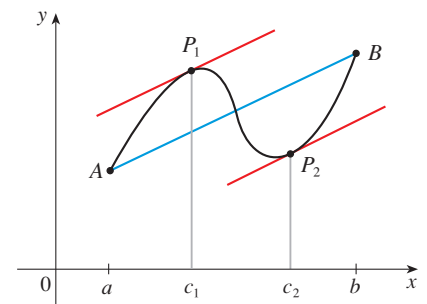


FIGURE 2

The slope of the secant line AB is

$$m_{AB} = \frac{f(b) - f(a)}{b - a}$$

which is the same expression as on the right side of Equation 1. Since $f'(c)$ is the slope of the tangent line at the point $(c, f(c))$, the Mean Value Theorem, in the form given by

Equation 1, says that there is at least one point $P(c, f(c))$ on the graph where the slope of the tangent line is the same as the slope of the secant line AB . In other words, there is a point P where the tangent line is parallel to the secant line AB . It seems clear that there is one such point P in Figure 1 and two such points P_1 and P_2 in Figure 2. (Imagine a line parallel to AB , starting far away and moving parallel to itself until it touches the graph for the first time.)

Because our intuition tells us that the Mean Value Theorem is true, we take it as the starting point for the development of the main facts of calculus. (When calculus is developed from first principles, however, the Mean Value Theorem is proved as a consequence of the axioms that define the real number system.)

EXAMPLE 1 | If an object moves in a straight line with position function $s = f(t)$, then the average velocity between $t = a$ and $t = b$ is

$$\frac{f(b) - f(a)}{b - a}$$

and the velocity at $t = c$ is $f'(c)$. Thus the Mean Value Theorem (in the form of Equation 1) tells us that at some time $t = c$ between a and b the instantaneous velocity $f'(c)$ is equal to that average velocity. For instance, if a car traveled 180 km in 2 hours, then the speedometer must have read 90 km/h at least once. ■

EXAMPLE 2 | **BB** **Compensatory growth** Experiments have been conducted in which individuals are deprived of food for a period of time during development and then placed back on a normal diet (see Figure 3). These experimental subjects display a period of reduced growth during the food deprivation, followed by a period of *compensatory growth* in which they catch up in size to individuals on a normal diet.

- Prove that there is always a time when an individual on a normal diet is growing at a rate equal to its average growth rate over the development period.
- Prove that there is always a time when an individual on a food-deprived diet is growing at a rate equal to its average growth rate over the development period.
- Given an example of each type of individual, show that there is a time after the point t_1 when food deprivation starts when the growth rate is nevertheless the same for both of them.

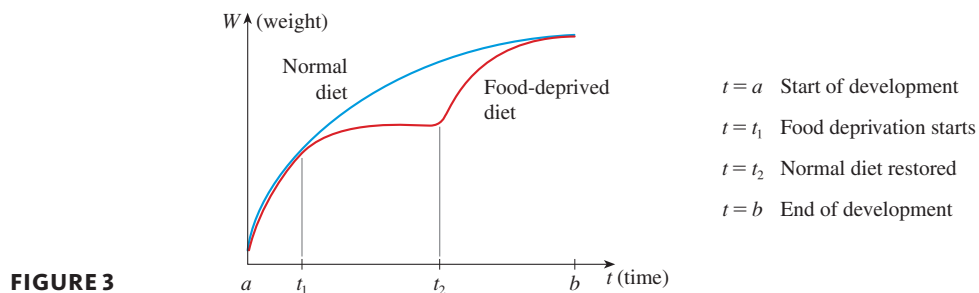


FIGURE 3

SOLUTION

- Let $W(t)$ be the weight of a specific individual on a normal diet at time t . Then W is a continuous function and is even differentiable. Let $t = a$ and $t = b$ be times at the beginning and end of the development period. By the Mean Value Theorem, there is a

number c such that

$$W'(c) = \frac{W(b) - W(a)}{b - a}$$

This equation says that at time $t = c$ the instantaneous rate of growth is equal to the average rate of growth.

(b) Exactly the same argument as in part (a) applies here. Because of the shape of the graph in this case, the time c will probably be different from the time in part (a).

(c) Let W_1 and W_2 be the weight functions for the normal and food-deprived individuals. Then the difference function $D(t) = W_1(t) - W_2(t)$ is differentiable and $D(t_1) = W_1(t_1) - W_2(t_1) = 0$ because the two individuals start with the same weight. Similarly, $D(b) = 0$. By the Mean Value Theorem there is a time c such that

$$D'(c) = \frac{D(b) - D(t_1)}{b - t_1} = \frac{0 - 0}{b - t_1} = 0$$

But $D'(c) = W_1'(c) - W_2'(c)$ and so $W_1'(c) = W_2'(c)$. So at time $t = c$ the two individuals were growing at the same rate. ■

The main significance of the Mean Value Theorem is that it enables us to obtain information about a function from information about its derivative. Our immediate use of this principle is to prove the basic facts concerning increasing and decreasing functions.

■ Increasing and Decreasing Functions

In Section 1.1 we defined increasing functions and decreasing functions and in Section 3.2 we observed from graphs that a function with a positive derivative is increasing. We now deduce this fact from the Mean Value Theorem.

Let's abbreviate the name of this test to the I/D Test.

Increasing/Decreasing Test

- (a) If $f'(x) > 0$ on an interval, then f is increasing on that interval.
- (b) If $f'(x) < 0$ on an interval, then f is decreasing on that interval.

PROOF

(a) Let x_1 and x_2 be any two numbers in the interval with $x_1 < x_2$. According to the definition of an increasing function (page 12), we have to show that $f(x_1) < f(x_2)$.

Because we are given that $f'(x) > 0$, we know that f is differentiable on $[x_1, x_2]$. So, by the Mean Value Theorem, there is a number c between x_1 and x_2 such that

$$(3) \quad f(x_2) - f(x_1) = f'(c)(x_2 - x_1)$$

Now $f'(c) > 0$ by assumption and $x_2 - x_1 > 0$ because $x_1 < x_2$. Thus the right side of Equation 3 is positive, and so

$$f(x_2) - f(x_1) > 0 \quad \text{or} \quad f(x_1) < f(x_2)$$

This shows that f is increasing.

Part (b) is proved similarly. ■

EXAMPLE 3 | Find where the function $f(x) = 3x^4 - 4x^3 - 12x^2 + 5$ is increasing and where it is decreasing.

SOLUTION First we calculate the derivative of f :

$$f'(x) = 12x^3 - 12x^2 - 24x = 12x(x - 2)(x + 1)$$

To use the I/D Test we have to know where $f'(x) > 0$ and where $f'(x) < 0$. This depends on the signs of the three factors of $f'(x)$, namely, $12x$, $x - 2$, and $x + 1$. We divide the real line into intervals whose endpoints are the critical numbers -1 , 0 , and 2 and arrange our work in a chart. A plus sign indicates that the given expression is positive, and a minus sign indicates that it is negative. The last column of the chart gives the conclusion based on the I/D Test. For instance, $f'(x) < 0$ for $0 < x < 2$, so f is decreasing on $(0, 2)$. (It would also be true to say that f is decreasing on the closed interval $[0, 2]$.)

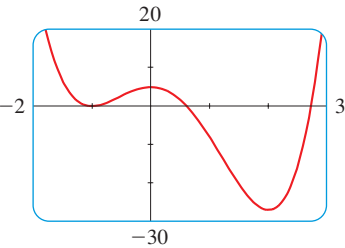


FIGURE 4

Interval	$12x$	$x - 2$	$x + 1$	$f'(x)$	f
$x < -1$	−	−	−	−	decreasing on $(-\infty, -1)$
$-1 < x < 0$	−	−	+	+	increasing on $(-1, 0)$
$0 < x < 2$	+	−	+	−	decreasing on $(0, 2)$
$x > 2$	+	+	+	+	increasing on $(2, \infty)$

The graph of f shown in Figure 4 confirms the information in the chart.

Recall from Section 4.1 that if f has a local maximum or minimum at c , then c must be a critical number of f (by Fermat’s Theorem), but not every critical number gives rise to a maximum or a minimum. We therefore need a test that will tell us whether or not f has a local maximum or minimum at a critical number.

You can see from Figure 4 that $f(0) = 5$ is a local maximum value of f because f increases on $(-1, 0)$ and decreases on $(0, 2)$. Or, in terms of derivatives, $f'(x) > 0$ for $-1 < x < 0$ and $f'(x) < 0$ for $0 < x < 2$. In other words, the sign of $f'(x)$ changes from positive to negative at 0 . This observation is the basis of the following test.

The First Derivative Test Suppose that c is a critical number of a continuous function f .

- (a) If f' changes from positive to negative at c , then f has a local maximum at c .
- (b) If f' changes from negative to positive at c , then f has a local minimum at c .
- (c) If f' does not change sign at c (for example, if f' is positive on both sides of c or negative on both sides), then f has no local maximum or minimum at c .

The First Derivative Test is a consequence of the I/D Test. In part (a), for instance, since the sign of $f'(x)$ changes from positive to negative at c , f is increasing to the left of c and decreasing to the right of c . It follows that f has a local maximum at c .

It is easy to remember the First Derivative Test by visualizing diagrams such as those in Figure 5.

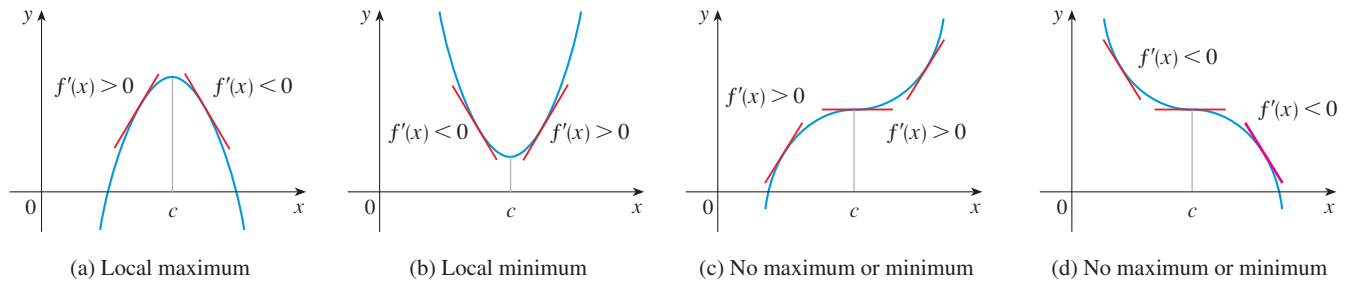


FIGURE 5

EXAMPLE 4 | Find the local minimum and maximum values of the function f in Example 3.

SOLUTION From the chart in the solution to Example 3 we see that $f'(x)$ changes from negative to positive at -1 , so $f(-1) = 0$ is a local minimum value by the First Derivative Test. Similarly, f' changes from negative to positive at 2 , so $f(2) = -27$ is also a local minimum value. As previously noted, $f(0) = 5$ is a local maximum value because $f'(x)$ changes from positive to negative at 0 .

Concavity

Let's see how the sign of $f''(x)$ affects the appearance of the graph of f . Since $f'' = (f')'$, we know that if $f''(x)$ is positive, then f' is an increasing function. This says that the slopes of the tangent lines of the curve $y = f(x)$ increase from left to right. Figure 6 shows the graph of such a function. The slope of this curve becomes progressively larger as x increases and we observe that, as a consequence, the curve bends upward. [It can be proved that this is equivalent to saying that the graph of f lies above all of its tangent lines.] Such a curve is called **concave upward**. In Figure 7, however, $f''(x)$ is negative, which means that f' is decreasing. Thus the slopes of f decrease from left to right and the curve bends downward. This curve is called **concave downward**. We summarize our discussion as follows.

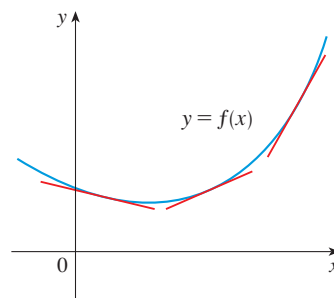


FIGURE 6
Since $f''(x) > 0$, the slopes increase and f is concave upward.

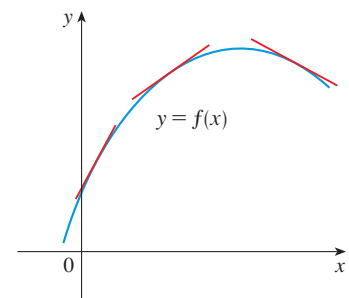


FIGURE 7
Since $f''(x) < 0$, the slopes decrease and f is concave downward.

Concavity Test

- (a) If $f''(x) > 0$ for all x in I , then the graph of f is concave upward on I .
- (b) If $f''(x) < 0$ for all x in I , then the graph of f is concave downward on I .

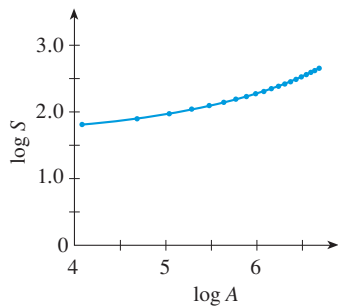


FIGURE 8

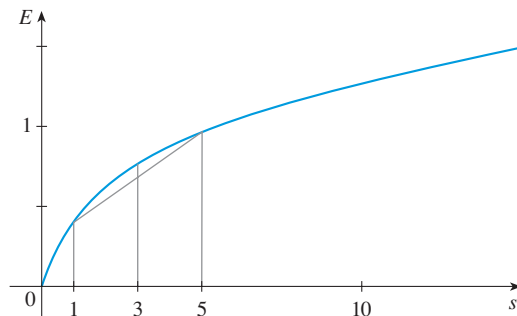
EXAMPLE 5 | **BB** **Species–area relationship** In Example 1.5.14 we explored the notion of species–area relationships by plotting the number of species S of bats against the areas A of their caves in central Mexico. We found that a log-log plot ($\log S$ as a function of $\log A$) was approximately linear and so a power model (for S as a function of A) was appropriate. We also found that a log-log plot was linear for the number of tree species in a given area of a rain forest in Malaysia (see Exercise 1.5.67). In both cases this means that, on a log scale, the rate of increase in number of species with an increase in area is the same no matter what the area is.

A recent study, however, has demonstrated that log-log plots are no longer linear when we consider areas that are substantially larger.¹ Figure 8 (from that study) shows a plot of $\log S$ versus $\log A$ for amphibians in Africa. You can see that $\log S$ is not a linear function of $\log A$ because its graph is concave upward. This means that, on a log scale, the rate of increase in the number of amphibian species with an increase in area is larger for big areas than for small ones. Put another way, the species–area relationship “accelerates” as the area increases.



FIGURE 9

EXAMPLE 6 | **BB** **Risk aversion in junco foraging**² Different junco habitats yield different amounts of seeds, and individuals can choose which habitat to feed in. The amount of energy reward E obtained from feeding in different habitats increases with the seed abundance s in the habitat but it does so at a decelerating rate (see Figure 9).



Suppose a bird can choose to feed exclusively in a habitat with $s = 3$ seeds per unit area or it can divide its time equally between two habitats with 1 and 5 seeds per unit area, respectively. For both choices the bird experiences an average of 3 seeds per unit area. Which choice provides the greatest energy reward?

SOLUTION The function $E(s)$ graphed in Figure 9 gives the energy reward as a function of seed density. Because the graph of E is concave downward, it lies below its tangent lines and above its secant lines. If we draw the secant line from $(1, E(1))$ to $(5, E(5))$, it will lie *below* the curve. The height of the secant line when $s = 3$ is the average of the heights when $s = 1$ and $s = 5$. Therefore

$$\frac{E(1) + E(5)}{2} < E(3)$$

1. D. Storch et al., “Universal Species–Area and Endemics–Area Relationships at Continental Scales,” *Nature* 488 (2012): 78–83.

2. T. Caraco et al., “An Empirical Demonstration of Risk-Sensitive Foraging Preferences,” *Animal Behavior* 28 (1980): 820–30.

This means that the junco gets more energy reward in a habitat with 3 seeds per unit area than it does by splitting its time between habitats with 1 and 5 seeds per unit area. ■

Figure 10 shows the graph of a function that is concave upward (abbreviated CU) on the intervals (b, c) , (d, e) , and (e, p) and concave downward (CD) on the intervals (a, b) , (c, d) , and (p, q) .

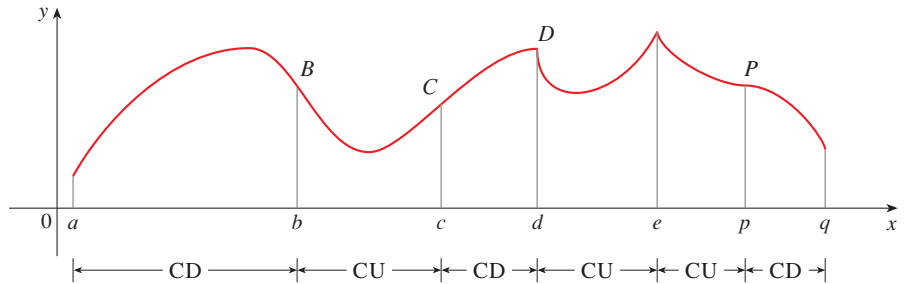


FIGURE 10

Notice in Figure 10 that the curve changes its direction of concavity when $x = b, c, d$, and p . The corresponding points on the curve (B, C, D , and P) are called **inflection points**. In general, a point P on a curve $y = f(x)$ is called an **inflection point** if f is continuous there and the curve changes from concave upward to concave downward or from concave downward to concave upward at P .

In view of the Concavity Test, there is a point of inflection at any point where the second derivative changes sign.

EXAMPLE 7 | Sketch a possible graph of a function f that satisfies the following conditions:

- (i) $f'(x) > 0$ on $(-\infty, 1)$, $f'(x) < 0$ on $(1, \infty)$
- (ii) $f''(x) > 0$ on $(-\infty, -2)$ and $(2, \infty)$, $f''(x) < 0$ on $(-2, 2)$
- (iii) $\lim_{x \rightarrow -\infty} f(x) = -2$, $\lim_{x \rightarrow \infty} f(x) = 0$

SOLUTION Condition (i) tells us that f is increasing on $(-\infty, 1)$ and decreasing on $(1, \infty)$. Condition (ii) says that f is concave upward on $(-\infty, -2)$ and $(2, \infty)$, and concave downward on $(-2, 2)$. From condition (iii) we know that the graph of f has two horizontal asymptotes: $y = -2$ and $y = 0$.

We first draw the horizontal asymptote $y = -2$ as a dashed line (see Figure 11). We then draw the graph of f approaching this asymptote at the far left, increasing to its maximum point at $x = 1$ and decreasing toward the x -axis as at the far right. We also make sure that the graph has inflection points when $x = -2$ and 2 . Notice that we made the curve bend upward for $x < -2$ and $x > 2$, and bend downward when x is between -2 and 2 . ■

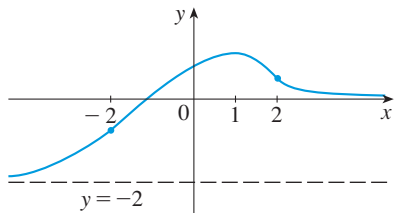


FIGURE 11

A consequence of the Concavity Test is the following test for maximum and minimum values.

The Second Derivative Test Suppose f'' is continuous near c .

- (a) If $f'(c) = 0$ and $f''(c) > 0$, then f has a local minimum at c .
- (b) If $f'(c) = 0$ and $f''(c) < 0$, then f has a local maximum at c .

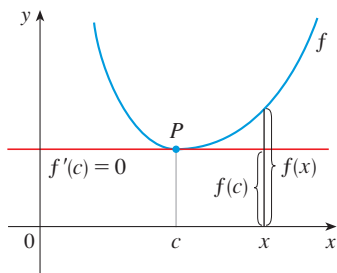


FIGURE 12
 $f''(x) > 0$, f is concave upward

For instance, part (a) is true because $f''(x) > 0$ near c and so f is concave upward near c . This means that the graph of f lies *above* its horizontal tangent at c and so f has a local minimum at c . (See Figure 12.)

EXAMPLE 8 | Discuss the curve $y = x^4 - 4x^3$ with respect to concavity, points of inflection, and local maxima and minima. Use this information to sketch the curve.

SOLUTION If $f(x) = x^4 - 4x^3$, then

$$f'(x) = 4x^3 - 12x^2 = 4x^2(x - 3)$$

$$f''(x) = 12x^2 - 24x = 12x(x - 2)$$

To find the critical numbers we set $f'(x) = 0$ and obtain $x = 0$ and $x = 3$. To use the Second Derivative Test we evaluate f'' at these critical numbers:

$$f''(0) = 0 \quad f''(3) = 36 > 0$$

Since $f'(3) = 0$ and $f''(3) > 0$, $f(3) = -27$ is a local minimum. Since $f''(0) = 0$, the Second Derivative Test gives no information about the critical number 0. But since $f'(x) < 0$ for $x < 0$ and also for $0 < x < 3$, the First Derivative Test tells us that f does not have a local maximum or minimum at 0.

Since $f''(x) = 0$ when $x = 0$ or 2, we divide the real line into intervals with these numbers as endpoints and complete the following chart.

Interval	$f''(x) = 12x(x - 2)$	Concavity
$(-\infty, 0)$	+	upward
$(0, 2)$	−	downward
$(2, \infty)$	+	upward

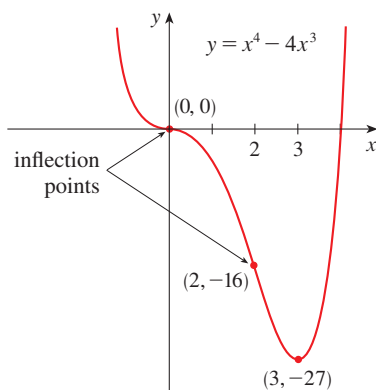


FIGURE 13

The point $(0, 0)$ is an inflection point since the curve changes from concave upward to concave downward there. Also $(2, -16)$ is an inflection point since the curve changes from concave downward to concave upward there.

Using the local minimum, the intervals of concavity, and the inflection points, we sketch the curve in Figure 13. ■

NOTE The Second Derivative Test is inconclusive when $f''(c) = 0$. In other words, at such a point there might be a maximum, there might be a minimum, or there might be neither (as in Example 8). This test also fails when $f''(c)$ does not exist. In such cases the First Derivative Test must be used. In fact, even when both tests apply, the First Derivative Test is often the easier one to use.

EXAMPLE 9 | Sketch the graph of the function $f(x) = x^{2/3}(6 - x)^{1/3}$.

SOLUTION Calculation of the first two derivatives gives

$$f'(x) = \frac{4 - x}{x^{1/3}(6 - x)^{2/3}} \quad f''(x) = \frac{-8}{x^{4/3}(6 - x)^{5/3}}$$

Use the differentiation rules to check these calculations.

Since $f'(x) = 0$ when $x = 4$ and $f'(x)$ does not exist when $x = 0$ or $x = 6$, the critical numbers are 0, 4, and 6.

Try reproducing the graph in Figure 14 with a graphing calculator or computer. Some machines produce the complete graph, some produce only the portion to the right of the y -axis, and some produce only the portion between $x = 0$ and $x = 6$. An equivalent expression that gives the correct graph is

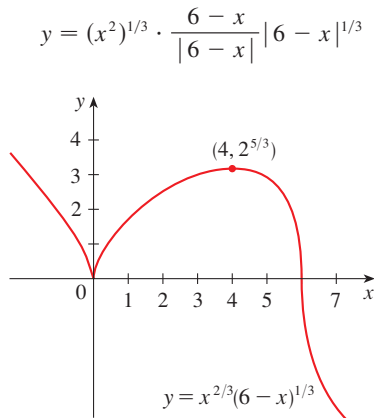


FIGURE 14

Interval	$4 - x$	$x^{1/3}$	$(6 - x)^{2/3}$	$f'(x)$	f
$x < 0$	+	−	+	−	decreasing on $(-\infty, 0)$
$0 < x < 4$	+	+	+	+	increasing on $(0, 4)$
$4 < x < 6$	−	+	+	−	decreasing on $(4, 6)$
$x > 6$	−	+	+	−	decreasing on $(6, \infty)$

To find the local extreme values we use the First Derivative Test. Because f' changes from negative to positive at 0, $f(0) = 0$ is a local minimum. Since f' changes from positive to negative at 4, $f(4) = 2^{5/3}$ is a local maximum. The sign of f' does not change at 6, so there is no minimum or maximum there. (The Second Derivative Test could be used at 4 but not at 0 or 6 because f'' does not exist at either of these numbers.)

Looking at the expression for $f''(x)$ and noting that $x^{4/3} \geq 0$ for all x , we have $f''(x) < 0$ for $x < 0$ and for $0 < x < 6$ and $f''(x) > 0$ for $x > 6$. So f is concave downward on $(-\infty, 0)$ and $(0, 6)$ and concave upward on $(6, \infty)$, and the only inflection point is $(6, 0)$. The graph is sketched in Figure 14. Note that the curve has vertical tangents at $(0, 0)$ and $(6, 0)$ because $|f'(x)| \rightarrow \infty$ as $x \rightarrow 0$ and as $x \rightarrow 6$.

EXAMPLE 10 | Bee population A population of honeybees raised in an apiary started with 50 bees at time $t = 0$ and was modeled by the function

$$P(t) = \frac{75,200}{1 + 1503e^{-0.5932t}}$$

where t is the time in weeks, $0 \leq t \leq 25$. Use a graph to estimate the time at which the bee population was growing fastest. Then use derivatives to give a more accurate estimate.

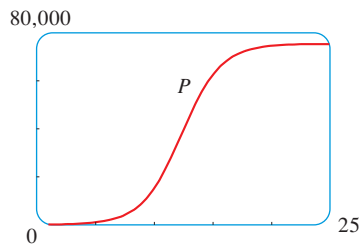


FIGURE 15

SOLUTION The population grows fastest when the population curve $y = P(t)$ has the steepest tangent line. From the graph of P in Figure 15, we estimate that the steepest tangent occurs when $t \approx 12$, so the bee population was growing most rapidly after about 12 weeks.

For a better estimate we calculate the derivative $P'(t)$, which is the rate of increase of the bee population:

$$P'(t) = \frac{67,046,785.92e^{-0.5932t}}{(1 + 1503e^{-0.5932t})^2}$$

We graph P' in Figure 16 and observe that P' has its maximum value when $t \approx 12.3$.

To get a still better estimate we note that f' has its maximum value when f' changes from increasing to decreasing. This happens when f changes from concave upward to concave downward, that is, when f has an inflection point. So we ask a CAS to compute the second derivative:

$$P''(t) \approx \frac{119,555,093,144e^{-1.1864t}}{(1 + 1503e^{-0.5932t})^3} - \frac{39,772,153e^{-0.5932t}}{(1 + 1503e^{-0.5932t})^2}$$

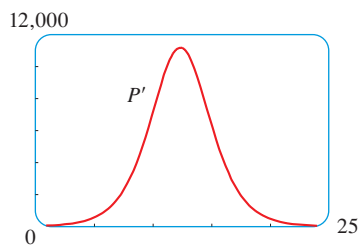


FIGURE 16

We could plot this function to see where it changes from positive to negative, but instead let's have the CAS solve the equation $P''(t) = 0$. It gives the answer $t \approx 12.3318$.

■ Graphing with Technology

When we use technology to graph a curve, our strategy is different from the one we've been using until now. Here we *start* with a graph produced by a graphing calculator or computer and then we refine it. We use calculus to make sure that we reveal all the important aspects of the curve. And with the use of graphing devices we can tackle curves that would be far too complicated to consider without technology.

EXAMPLE 11 | Graph the polynomial $f(x) = 2x^6 + 3x^5 + 3x^3 - 2x^2$. Use the graphs of f' and f'' to estimate all maximum and minimum points and intervals of concavity.

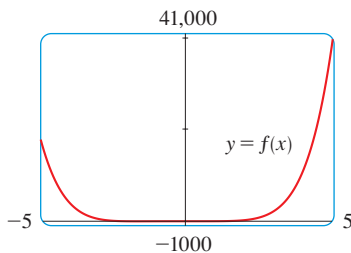


FIGURE 17

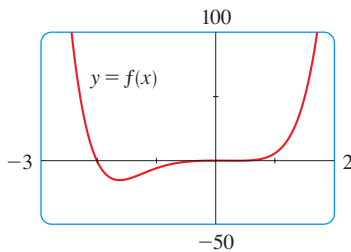


FIGURE 18

SOLUTION If we specify a domain but not a range, many graphing devices will deduce a suitable range from the values computed. Figure 17 shows the plot from one such device if we specify that $-5 \leq x \leq 5$. Although this viewing rectangle is useful for showing that the asymptotic behavior (or end behavior) is the same as for $y = 2x^6$, it is obviously hiding some finer detail. So we change to the viewing rectangle $[-3, 2]$ by $[-50, 100]$ shown in Figure 18.

From this graph it appears that there is an absolute minimum value of about -15.33 when $x \approx -1.62$ (by using the cursor) and f is decreasing on $(-\infty, -1.62)$ and increasing on $(-1.62, \infty)$. Also there appears to be a horizontal tangent at the origin and inflection points when $x = 0$ and when x is somewhere between -2 and -1 .

Now let's try to confirm these impressions using calculus. We differentiate and get

$$f'(x) = 12x^5 + 15x^4 + 9x^2 - 4x$$

$$f''(x) = 60x^4 + 60x^3 + 18x - 4$$

When we graph f' in Figure 19 we see that $f'(x)$ changes from negative to positive when $x \approx -1.62$; this confirms (by the First Derivative Test) the minimum value that we found earlier. But, perhaps to our surprise, we also notice that $f'(x)$ changes from positive to negative when $x = 0$ and from negative to positive when $x \approx 0.35$. This means that f has a local maximum at 0 and a local minimum when $x \approx 0.35$, but these were hidden in Figure 18. Indeed, if we now zoom in toward the origin in Figure 20, we see what we missed before: a local maximum value of 0 when $x = 0$ and a local minimum value of about -0.1 when $x \approx 0.35$.

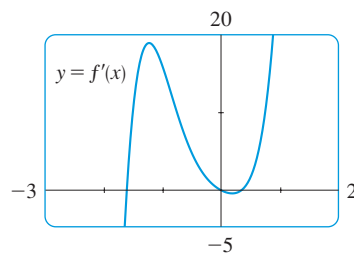


FIGURE 19

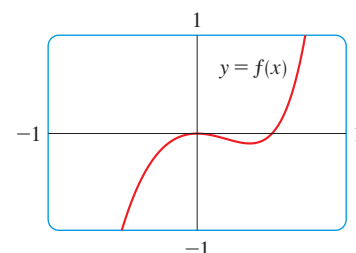


FIGURE 20

What about concavity and inflection points? From Figures 18 and 20 there appear to be inflection points when x is a little to the left of -1 and when x is a little to the right of 0. But it's difficult to determine inflection points from the graph of f , so we graph

the second derivative f'' in Figure 21. We see that f'' changes from positive to negative when $x \approx -1.23$ and from negative to positive when $x \approx 0.19$. So, correct to two decimal places, f is concave upward on $(-\infty, -1.23)$ and $(0.19, \infty)$ and concave downward on $(-1.23, 0.19)$. The inflection points are $(-1.23, -10.18)$ and $(0.19, -0.05)$.

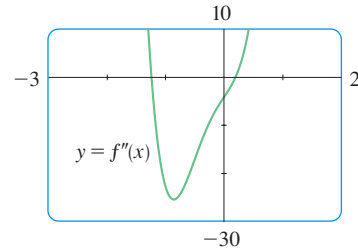
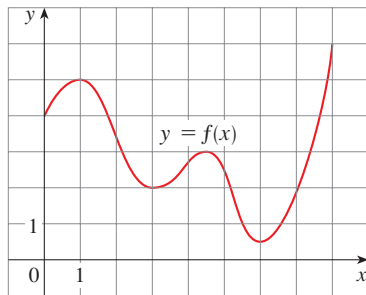


FIGURE 21

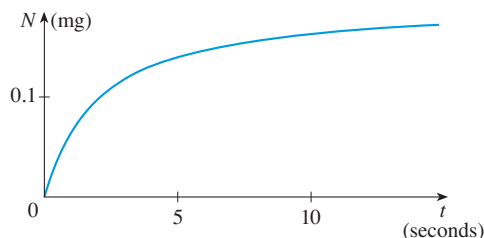
We have discovered that no single graph reveals all the important features of this polynomial. But Figures 18 and 20, when taken together, do provide an accurate picture.

EXERCISES 4.2

- Use the graph of f to estimate the values of c that satisfy the conclusion of the Mean Value Theorem for the interval $[0, 8]$.



- Foraging** Many animals forage on resources that are distributed in discrete patches. For example, bumblebees visit many flowers, foraging on nectar from each. The amount of nectar $N(t)$ consumed from any flower increases with the amount of time spent at that flower, but with diminishing returns, as illustrated.

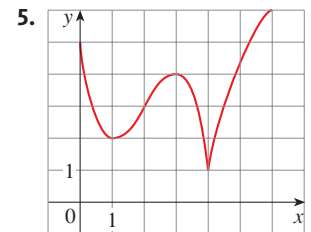
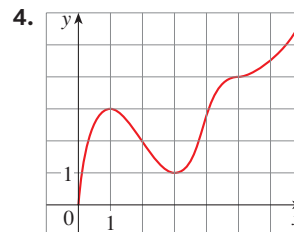


- What does this mean about the first and second derivatives of N ?

- What is the average rate at which nectar is consumed over the first 10 seconds?
 - The Mean Value Theorem tells us that there exists a time at which the instantaneous rate of nectar consumption is equal to the average found in part (b). Illustrate this idea graphically and estimate the time at which this occurs.
- Suppose that $3 \leq f'(x) \leq 5$ for all values of x . Show that $18 \leq f(8) - f(2) \leq 30$.

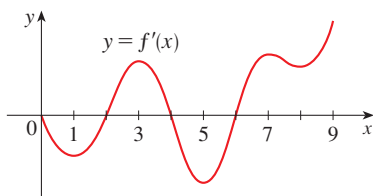
4–5 Use the given graph of f to find the following.

- The open intervals on which f is increasing.
- The open intervals on which f is decreasing.
- The open intervals on which f is concave upward.
- The open intervals on which f is concave downward.
- The coordinates of the points of inflection.

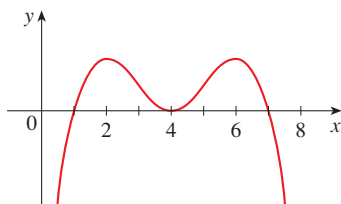


- Suppose you are given a formula for a function f .
 - How do you determine where f is increasing or decreasing?
 - How do you determine where the graph of f is concave upward or concave downward?
 - How do you locate inflection points?

7. (a) State the First Derivative Test.
 (b) State the Second Derivative Test. Under what circumstances is it inconclusive? What do you do if it fails?
8. The graph of the first derivative f' of a function f is shown.
 (a) On what intervals is f increasing? Explain.
 (b) At what values of x does f have a local maximum or minimum? Explain.
 (c) On what intervals is f concave upward or concave downward? Explain.
 (d) What are the x -coordinates of the inflection points of f ? Why?



9. In each part state the x -coordinates of the inflection points of f . Give reasons for your answers.
 (a) The curve is the graph of f .
 (b) The curve is the graph of f' .
 (c) The curve is the graph of f'' .



10. **HIV prevalence** The table gives the number of HIV-infected men in San Francisco from 1982 to 1991.

Year	Number of infections	Year	Number of infections
1982	80	1987	3500
1983	300	1988	4500
1984	700	1989	6000
1985	1500	1990	7200
1986	2500	1991	9000

- (a) If $H(t)$ is the number of infected men at time t , plot the values of $H(t)$. What does the direction of concavity appear to be? Provide a biological interpretation.
 (b) Use the table to construct a table of estimated values for $H'(t)$.
 (c) Use the table of values of $H'(t)$ in part (b) to construct a table of values for $H''(t)$. Do the values corroborate your answer to part (a)?

11–20

- (a) Find the intervals on which f is increasing or decreasing.
 (b) Find the local maximum and minimum values of f .
 (c) Find the intervals of concavity and the inflection points.

11. $f(x) = 2x^3 + 3x^2 - 36x$

12. $f(x) = 4x^3 + 3x^2 - 6x + 1$

13. $f(x) = x^4 - 2x^2 + 3$

14. $f(x) = \frac{x^2}{x^2 + 3}$

15. $f(x) = \sin x + \cos x, \quad 0 \leq x \leq 2\pi$

16. $f(x) = \cos^2 x - 2 \sin x, \quad 0 \leq x \leq 2\pi$

17. $f(x) = e^{2x} + e^{-x}$

18. $f(x) = x^2 \ln x$

19. $f(x) = (\ln x)/\sqrt{x}$

20. $f(x) = \sqrt{x}e^{-x}$

- 21–22 Find the local maximum and minimum values of f using both the First and Second Derivative Tests. Which method do you prefer?

21. $f(x) = x + \sqrt{1-x}$

22. $f(x) = \frac{x}{x^2 + 4}$

23. Suppose f'' is continuous on $(-\infty, \infty)$.

- (a) If $f'(2) = 0$ and $f''(2) = -5$, what can you say about f ?
 (b) If $f'(6) = 0$ and $f''(6) = 0$, what can you say about f ?

24. (a) Find the critical numbers of $f(x) = x^4(x-1)^3$.
 (b) What does the Second Derivative Test tell you about the behavior of f at these critical numbers?
 (c) What does the First Derivative Test tell you?

25–36

- (a) Find the intervals of increase or decrease.
 (b) Find the local maximum and minimum values.
 (c) Find the intervals of concavity and the inflection points.
 (d) Use the information from parts (a)–(c) to sketch the graph.
 Check your work with a graphing device if you have one.

25. $f(x) = 2x^3 - 3x^2 - 12x$

26. $f(x) = 2 + 3x - x^3$

27. $f(x) = 2 + 2x^2 - x^4$

28. $g(x) = 200 + 8x^3 + x^4$

29. $h(x) = (x+1)^5 - 5x - 2$

30. $h(x) = x^5 - 2x^3 + x$

31. $A(x) = x\sqrt{x+3}$

32. $B(x) = 3x^{2/3} - x$

33. $C(x) = x^{1/3}(x+4)$

34. $f(x) = \ln(x^4 + 27)$

35. $f(\theta) = 2 \cos \theta + \cos^2 \theta, \quad 0 \leq \theta \leq 2\pi$

36. $f(t) = t + \cos t, \quad -2\pi \leq t \leq 2\pi$

37–44

- (a) Find the vertical and horizontal asymptotes.
 (b) Find the intervals of increase or decrease.
 (c) Find the local maximum and minimum values.
 (d) Find the intervals of concavity and the inflection points.
 (e) Use the information from parts (a)–(d) to sketch the graph of f .

37. $f(x) = \frac{x^2}{x^2 - 1}$

38. $f(x) = \frac{x^2}{(x - 2)^2}$

39. $f(x) = \sqrt{x^2 + 1} - x$

40. $f(x) = x \tan x, \quad -\pi/2 < x < \pi/2$

41. $f(x) = \ln(1 - \ln x)$

42. $f(x) = \frac{e^x}{1 + e^x}$

43. $f(x) = e^{-1/(x+1)}$

44. $f(x) = e^{\arctan x}$

45. Suppose the derivative of a function f is $f'(x) = (x + 1)^2(x - 3)^5(x - 6)^4$. On what interval is f increasing?
46. Use the methods of this section to sketch the curve $y = x^3 - 3a^2x + 2a^3$, where a is a positive constant. What do the members of this family of curves have in common? How do they differ from each other?
47. Let $f(t)$ be the temperature at time t where you live and suppose that at time $t = 3$ you feel uncomfortably hot. How do you feel about the given data in each case?
- (a) $f'(3) = 2, \quad f''(3) = 4$
 (b) $f'(3) = 2, \quad f''(3) = -4$
 (c) $f'(3) = -2, \quad f''(3) = 4$
 (d) $f'(3) = -2, \quad f''(3) = -4$
48. Suppose $f(3) = 2, f'(3) = \frac{1}{2}$, and $f'(x) > 0$ and $f''(x) < 0$ for all x .
- (a) Sketch a possible graph for f .
 (b) How many solutions does the equation $f(x) = 0$ have? Why?
 (c) Is it possible that $f'(2) = \frac{1}{3}$? Why?
49. Coffee is being poured into the mug shown in the figure at a constant rate (measured in volume per unit time). Sketch a rough graph of the depth of the coffee in the mug as a function of time. Account for the shape of the graph in terms of concavity. What is the significance of the inflection point?



50. **Antibiotic pharmacokinetics** Suppose that antibiotics are injected into a patient to treat a sinus infection. The antibiotics circulate in the blood, slowly diffusing into the sinus cavity while simultaneously being filtered out of the blood by the liver. In Chapter 10 we will derive a model for the concentration of the antibiotic in the sinus cavity as a function of time since the injection:

$$c(t) = \frac{e^{-\alpha t} - e^{-\beta t}}{\beta - \alpha} \quad \text{where } \beta > \alpha > 0$$

- (a) At what time does c have its maximum value?
 (b) At what time does the inflection point occur? What is the significance of the inflection point for the concentration function?
 (c) Sketch the graph of c .
51. A **drug-loading curve** describes the level of medication in the bloodstream after a drug is administered. A *surge function* $S(t) = At^p e^{-kt}$ is often used to model the loading curve, reflecting an initial surge in the drug level and then a more gradual decline. If, for a particular drug, $A = 0.01$, $p = 4$, $k = 0.07$, and t is measured in minutes, estimate the times corresponding to the inflection points and explain their significance. If you have a graphing device, use it to graph the drug response curve.
52. **Mutation accumulation** When a population is subjected to a mutagen, the fraction of the population that contains at least one mutation increases with the duration of the exposure. A commonly used equation describing this fraction is $f(t) = 1 - e^{-\mu t}$, where μ is the mutation rate and is positive. Suppose we have two populations, A and B. Population A is subjected to the mutagen for 3 min whereas, with population B, half of the individuals are subjected to the mutagen for 2 min and the other half for 4 min. Which population will have the largest fraction of mutants? Explain your answer using derivatives.
53. A **dose response curve** in pharmacology is a plot of the effectiveness R of a drug as a function of the drug concentration c . Such curves typically increase with an S-shape, a simple mathematical model being

$$R(c) = \frac{c^2}{3 + c^2}$$

- (a) At what drug concentration does the inflection point occur?
 (b) Suppose we have two different treatment protocols, one where the concentration is held steady at 2 and another in which the concentration varies through time, spending equal amounts of time at 1.5 and 2.5. Which protocol would have the greater response?
54. The family of bell-shaped curves

$$y = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)}$$

occurs in probability and statistics, where it is called the *normal density function*. The constant μ is called the *mean* and the positive constant σ is called the *standard deviation*. For simplicity, let's scale the function so as to remove the factor $1/(\sigma\sqrt{2\pi})$ and let's analyze the special case where $\mu = 0$. So we study the function

$$f(x) = e^{-x^2/(2\sigma^2)}$$

- (a) Find the asymptote, maximum value, and inflection points of f .
 (b) What role does σ play in the shape of the curve?
 (c) Illustrate by graphing four members of this family on the same screen.



55. In the theory of relativity, the mass of a particle is

$$m = \frac{m_0}{\sqrt{1 - v^2/c^2}}$$

where m_0 is the rest mass of the particle, m is the mass when the particle moves with speed v relative to the observer, and c is the speed of light. Sketch the graph of m as a function of v .

56. In the theory of relativity, the energy of a particle is

$$E = \sqrt{m_0^2 c^4 + h^2 c^2 / \lambda^2}$$

where m_0 is the rest mass of the particle, λ is its wave length, and h is Planck's constant. Sketch the graph of E as a function of λ . What does the graph say about the energy?

57. Find a cubic function $f(x) = ax^3 + bx^2 + cx + d$ that has a local maximum value of 3 at $x = -2$ and a local minimum value of 0 at $x = 1$.

58. For what values of the numbers a and b does the function

$$f(x) = axe^{bx^2}$$

have the maximum value $f(2) = 1$?



59–62 Produce graphs of f that reveal all the important aspects of the curve. In particular, you should use graphs of f' and f'' to estimate the intervals of increase and decrease, extreme values, intervals of concavity, and inflection points.

59. $f(x) = 4x^4 - 32x^3 + 89x^2 - 95x + 29$

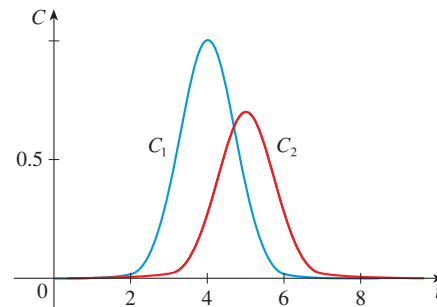
60. $f(x) = x^6 - 15x^5 + 75x^4 - 125x^3 - x$

61. $f(x) = x^2 - 4x + 7 \cos x, \quad -4 \leq x \leq 4$

62. $f(x) = \tan x + 5 \cos x$

63. **Growth rate** A 20-year-old university student weighs 138 lb and had a birth weight of 6 lb. Prove that at some point in her life she was growing at a rate of 6.6 pounds per year.

64. **Antibiotic concentration** Suppose an antibiotic is administered orally. It is first absorbed into the bloodstream, from which it passes into the sinus cavity. It is also metabolized from both sites. The concentrations $C_1(t)$ in the blood and $C_2(t)$ in the sinus cavity are shown. Prove that there is a time when the concentration in each site is increasing at the same rate. Prove that there is also a time when the concentration in each site is decreasing at the same rate.



65. Show that a cubic function (a third-degree polynomial) always has exactly one point of inflection. If its graph has three x -intercepts x_1 , x_2 , and x_3 , show that the x -coordinate of the inflection point is $(x_1 + x_2 + x_3)/3$.



66. For what values of c does the polynomial $P(x) = x^4 + cx^3 + x^2$ have two inflection points? One inflection point? None? Illustrate by graphing P for several values of c . How does the graph change as c decreases?

4.3 L'Hospital's Rule: Comparing Rates of Growth

■ Indeterminate Quotients

Suppose we are trying to analyze the behavior of the function

$$F(x) = \frac{\ln x}{x - 1}$$

Although F is not defined when $x = 1$, we need to know how F behaves *near* 1. In

particular, we would like to know the value of the limit

$$(1) \quad \lim_{x \rightarrow 1} \frac{\ln x}{x - 1}$$

In computing this limit we can't apply Law 5 of limits (the limit of a quotient is the quotient of the limits, see Section 2.4) because the limit of the denominator is 0. In fact, although the limit in (1) exists, its value is not obvious because both numerator and denominator approach 0 and $\frac{0}{0}$ is not defined.

In general, if we have a limit of the form

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$$

where both $f(x) \rightarrow 0$ and $g(x) \rightarrow 0$ as $x \rightarrow a$, then this limit may or may not exist and is called an **indeterminate form of type $\frac{0}{0}$** . We met some limits of this type in Chapter 2. For rational functions, we can cancel common factors:

$$\lim_{x \rightarrow 1} \frac{x^2 - x}{x^2 - 1} = \lim_{x \rightarrow 1} \frac{x(x - 1)}{(x + 1)(x - 1)} = \lim_{x \rightarrow 1} \frac{x}{x + 1} = \frac{1}{2}$$

We used a geometric argument to show that

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

But these methods do not work for limits such as (1), so in this section we introduce a systematic method, known as *l'Hospital's Rule*, for the evaluation of indeterminate forms.

Another situation in which a limit is not obvious occurs when we look for a horizontal asymptote of F and need to evaluate the limit

$$(2) \quad \lim_{x \rightarrow \infty} \frac{\ln x}{x - 1}$$

It isn't obvious how to evaluate this limit because both numerator and denominator become large as $x \rightarrow \infty$. There is a struggle between numerator and denominator. If the numerator wins, the limit will be ∞ ; if the denominator wins, the answer will be 0. Or there may be some compromise, in which case the answer will be some finite positive number.

In general, if we have a limit of the form

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$$

where both $f(x) \rightarrow \infty$ (or $-\infty$) and $g(x) \rightarrow \infty$ (or $-\infty$), then the limit may or may not exist and is called an **indeterminate form of type ∞/∞** . We saw in Section 2.2 that this type of limit can be evaluated for certain functions, including rational functions, by dividing numerator and denominator by the highest power of x that occurs in the denominator. For instance,

$$\lim_{x \rightarrow \infty} \frac{x^2 - 1}{2x^2 + 1} = \lim_{x \rightarrow \infty} \frac{1 - \frac{1}{x^2}}{2 + \frac{1}{x^2}} = \frac{1 - 0}{2 + 0} = \frac{1}{2}$$

L'Hospital

L'Hospital's Rule is named after a French nobleman, the Marquis de l'Hospital (1661–1704), but was discovered by a Swiss mathematician, John Bernoulli (1667–1748). You might sometimes see l'Hospital spelled as l'Hôpital, but he spelled his own name l'Hospital, as was common in the 17th century. See Exercise 57 for the example that the Marquis used to illustrate his rule.

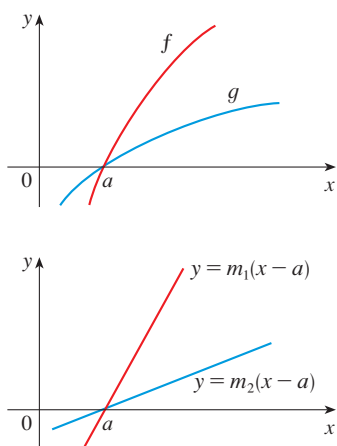


FIGURE 1

Figure 1 suggests visually why l'Hospital's Rule might be true. The first graph shows two differentiable functions f and g , each of which approaches 0 as $x \rightarrow a$. If we were to zoom in toward the point $(a, 0)$, the graphs would start to look almost linear. But if the functions actually were linear, as in the second graph, then their ratio would be

$$\frac{m_1(x - a)}{m_2(x - a)} = \frac{m_1}{m_2}$$

which is the ratio of their derivatives. This suggests that

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

This method does not work for limits such as (2), but l'Hospital's Rule also applies to this type of indeterminate form.

L'Hospital's Rule Suppose f and g are differentiable and $g'(x) \neq 0$ near a (except possibly at a). Suppose that

$$\lim_{x \rightarrow a} f(x) = 0 \quad \text{and} \quad \lim_{x \rightarrow a} g(x) = 0$$

$$\text{or that} \quad \lim_{x \rightarrow a} f(x) = \pm\infty \quad \text{and} \quad \lim_{x \rightarrow a} g(x) = \pm\infty$$

(In other words, we have an indeterminate form of type $\frac{0}{0}$ or ∞/∞ .) Then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

if the limit on the right side exists (or is ∞ or $-\infty$).

NOTE 1 L'Hospital's Rule says that the limit of a quotient of functions is equal to the limit of the quotient of their derivatives, provided that the given conditions are satisfied. It is especially important to verify the conditions regarding the limits of f and g before using l'Hospital's Rule.

NOTE 2 L'Hospital's Rule is also valid for one-sided limits and for limits at infinity or negative infinity; that is, " $x \rightarrow a$ " can be replaced by any of the symbols $x \rightarrow a^+$, $x \rightarrow a^-$, $x \rightarrow \infty$, or $x \rightarrow -\infty$.

NOTE 3 For the special case in which $f(a) = g(a) = 0$, f' and g' are continuous, and $g'(a) \neq 0$, it is easy to see why l'Hospital's Rule is true. In fact, using the alternative form of the definition of a derivative, we have

$$\begin{aligned} \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} &= \frac{f'(a)}{g'(a)} = \frac{\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}}{\lim_{x \rightarrow a} \frac{g(x) - g(a)}{x - a}} = \lim_{x \rightarrow a} \frac{\frac{f(x) - f(a)}{x - a}}{\frac{g(x) - g(a)}{x - a}} \\ &= \lim_{x \rightarrow a} \frac{f(x) - f(a)}{g(x) - g(a)} = \lim_{x \rightarrow a} \frac{f(x)}{g(x)} \end{aligned}$$

The general version of l'Hospital's Rule is more difficult; its proof can be found in more advanced books.

EXAMPLE 1 Find $\lim_{x \rightarrow 1} \frac{\ln x}{x - 1}$.

SOLUTION Since

$$\lim_{x \rightarrow 1} \ln x = \ln 1 = 0 \quad \text{and} \quad \lim_{x \rightarrow 1} (x - 1) = 0$$

we can apply l'Hospital's Rule:

 Notice that when using l'Hospital's Rule we differentiate the numerator and denominator *separately*. We do *not* use the Quotient Rule.

$$\begin{aligned}\lim_{x \rightarrow 1} \frac{\ln x}{x - 1} &= \lim_{x \rightarrow 1} \frac{\frac{d}{dx}(\ln x)}{\frac{d}{dx}(x - 1)} = \lim_{x \rightarrow 1} \frac{1/x}{1} \\ &= \lim_{x \rightarrow 1} \frac{1}{x} = 1\end{aligned}$$

The graph of the function of Example 2 is shown in Figure 2. We have noticed previously that exponential functions grow far more rapidly than power functions, so the result of Example 2 is not unexpected. See also Exercise 55.

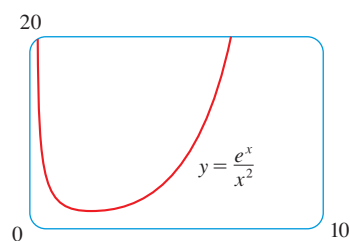


FIGURE 2

The graph of the function of Example 3 is shown in Figure 3. We have discussed previously the slow growth of logarithms, so it isn't surprising that this ratio approaches 0 as $x \rightarrow \infty$. See also Exercise 56.

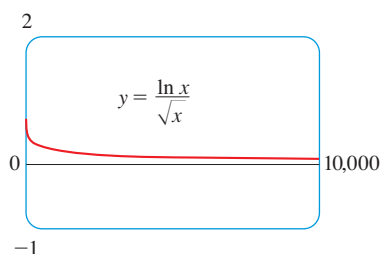


FIGURE 3

EXAMPLE 2 | Calculate $\lim_{x \rightarrow \infty} \frac{e^x}{x^2}$.

SOLUTION We have $\lim_{x \rightarrow \infty} e^x = \infty$ and $\lim_{x \rightarrow \infty} x^2 = \infty$, so l'Hospital's Rule gives

$$\lim_{x \rightarrow \infty} \frac{e^x}{x^2} = \lim_{x \rightarrow \infty} \frac{\frac{d}{dx}(e^x)}{\frac{d}{dx}(x^2)} = \lim_{x \rightarrow \infty} \frac{e^x}{2x}$$

Since $e^x \rightarrow \infty$ and $2x \rightarrow \infty$ as $x \rightarrow \infty$, the limit on the right side is also indeterminate, but a second application of l'Hospital's Rule gives

$$\lim_{x \rightarrow \infty} \frac{e^x}{x^2} = \lim_{x \rightarrow \infty} \frac{e^x}{2x} = \lim_{x \rightarrow \infty} \frac{e^x}{2} = \infty$$

EXAMPLE 3 | Calculate $\lim_{x \rightarrow \infty} \frac{\ln x}{\sqrt{x}}$.

SOLUTION Since $\ln x \rightarrow \infty$ and $\sqrt{x} \rightarrow \infty$ as $x \rightarrow \infty$, l'Hospital's Rule applies:

$$\lim_{x \rightarrow \infty} \frac{\ln x}{\sqrt{x}} = \lim_{x \rightarrow \infty} \frac{1/x}{\frac{1}{2}x^{-1/2}}$$

Notice that the limit on the right side is now indeterminate of type $\frac{0}{0}$. But instead of applying l'Hospital's Rule a second time as we did in Example 2, we simplify the expression and see that a second application is unnecessary:

$$\lim_{x \rightarrow \infty} \frac{\ln x}{\sqrt{x}} = \lim_{x \rightarrow \infty} \frac{1/x}{\frac{1}{2}x^{-1/2}} = \lim_{x \rightarrow \infty} \frac{2}{\sqrt{x}} = 0$$

EXAMPLE 4 | **Glucose administration** In Exercise 7.4.43 you are asked to derive an equation for the concentration of glucose in the bloodstream during intravenous injection. For a specific choice of constants, the concentration one minute after injection begins is

$$C = \frac{r}{k}(1 - e^{-k})$$

where r is the injection rate and k is the rate at which glucose is metabolized from the blood. Different patients metabolize glucose at different rates, meaning they have different values of k . You can verify that C is a decreasing function of k and therefore, by taking the limit as $k \rightarrow 0$, we obtain an upper bound for the predicted concentration of any patient. Find this limit.

SOLUTION Because $k \rightarrow 0$ and $1 - e^{-k} \rightarrow 0$, we can apply l'Hospital's Rule:

$$\begin{aligned}\lim_{k \rightarrow 0} C &= \lim_{k \rightarrow 0} r \cdot \frac{1 - e^{-k}}{k} = r \lim_{k \rightarrow 0} \frac{e^{-k}}{1} \\ &= r \cdot 1 = r\end{aligned}$$

Thus the upper bound for the concentration is r , the same as the injection rate. ■

EXAMPLE 5 | Find $\lim_{x \rightarrow \pi^-} \frac{\sin x}{1 - \cos x}$.

SOLUTION If we blindly attempted to use l'Hospital's Rule, we would get

$$\lim_{x \rightarrow \pi^-} \frac{\sin x}{1 - \cos x} = \lim_{x \rightarrow \pi^-} \frac{\cos x}{\sin x} = -\infty$$

✗ This is **wrong!** Although the numerator $\sin x \rightarrow 0$ as $x \rightarrow \pi^-$, notice that the denominator $(1 - \cos x)$ does not approach 0, so l'Hospital's Rule can't be applied here.

The required limit is, in fact, easy to find because the function is continuous at π and the denominator is nonzero there:

$$\lim_{x \rightarrow \pi^-} \frac{\sin x}{1 - \cos x} = \frac{\sin \pi}{1 - \cos \pi} = \frac{0}{1 - (-1)} = 0$$
 ■

Example 5 shows what can go wrong if you use l'Hospital's Rule without thinking. Other limits *can* be found using l'Hospital's Rule but are more easily found by other methods. (See Examples 2.4.3, 2.4.5, and 2.2.5 and the discussion at the beginning of this section.) So when evaluating any limit, you should consider other methods before using l'Hospital's Rule.

■ Which Functions Grow Fastest?

L'Hospital's Rule enables us to compare the rates of growth of functions. Suppose we have two functions $f(x)$ and $g(x)$ that both become large as x becomes large, that is,

$$\lim_{x \rightarrow \infty} f(x) = \infty \quad \text{and} \quad \lim_{x \rightarrow \infty} g(x) = \infty$$

We say that $f(x)$ approaches infinity **more quickly** than $g(x)$ if

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \infty$$

and that $f(x)$ approaches infinity **more slowly** than $g(x)$ if

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = 0$$

For example, we used l'Hospital's Rule in Example 2 to show that

$$\lim_{x \rightarrow \infty} \frac{e^x}{x^2} = \infty$$

and so the exponential function $y = e^x$ grows more quickly than $y = x^2$. In fact $y = e^x$

grows more quickly than all the power functions $y = x^n$ (see Exercise 55). This fact is illustrated in Figure 4, where you can see that $y = x^3$ exceeds $y = e^x$ some of the time but after about $x = 4.5$ the exponential function overtakes the other functions.

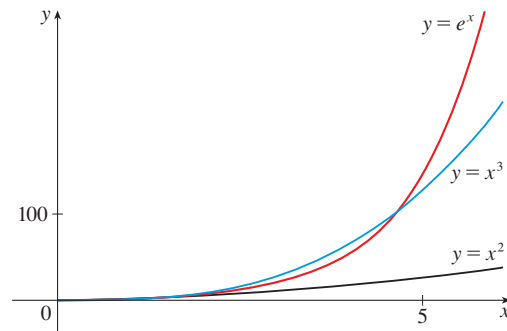


FIGURE 4

EXAMPLE 6 | Rank the following functions in order of how quickly they approach infinity as $x \rightarrow \infty$:

$$y = \ln x \quad y = x \quad y = e^{0.1x} \quad y = \sqrt{x}$$

SOLUTION If we try to decide the ranking by plotting the four functions as in Figure 5(a), we get a misleading picture: it looks as if $y = x$ is the winner. But l'Hospital's Rule tells us that can't be true:

$$\lim_{x \rightarrow \infty} \frac{x}{e^{0.1x}} = \lim_{x \rightarrow \infty} \frac{1}{0.1e^{0.1x}} = 0$$

so $y = x$ grows more slowly than $y = e^{0.1x}$.

We know from Example 3 that $y = \ln x$ grows more slowly than $y = \sqrt{x}$. Also, $y = \sqrt{x}$ grows more slowly than $y = x$ because

$$\frac{\sqrt{x}}{x} = \frac{1}{\sqrt{x}} \rightarrow 0 \quad \text{as } x \rightarrow \infty$$

So the ranking, from fastest to slowest, is as follows:

$$y = e^{0.1x} \quad y = x \quad y = \sqrt{x} \quad y = \ln x$$

This ranking is illustrated in Figure 5(b).

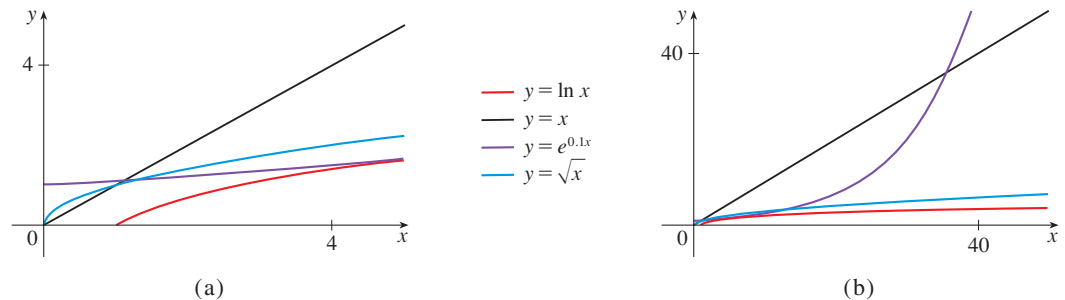


FIGURE 5

EXAMPLE 7 | Analyze the limiting behavior of the function

$$f(x) = \frac{x + x^4}{1 + x^2 + e^x}$$

by considering just the dominant terms in the numerator and denominator.

SOLUTION The dominant term in the numerator is x^4 because it becomes large more quickly than x . We know from Example 2 that e^x is the dominant term in the denominator. If we retain just the two dominant terms, we get the simpler function

$$g(x) = \frac{x^4}{e^x}$$

The exponential function grows faster than any power of x and so $\lim_{x \rightarrow \infty} g(x) = 0$. Therefore we expect that $\lim_{x \rightarrow \infty} f(x) = 0$ too. This could be verified directly with four applications of l'Hospital's Rule, but the important thing is to recognize that our intuition about orders of magnitude leads us to the correct answer. In Figure 6 we see that the functions f and g have practically identical behavior for large values of x . ■

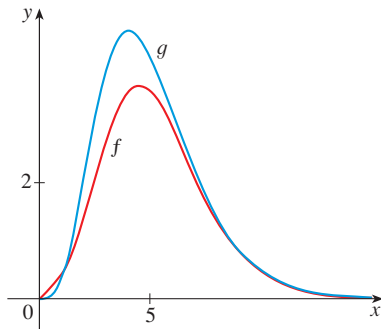


FIGURE 6

Indeterminate Products

If $\lim_{x \rightarrow a} f(x) = 0$ and $\lim_{x \rightarrow a} g(x) = \infty$ (or $-\infty$), then it isn't clear what the value of $\lim_{x \rightarrow a} f(x)g(x)$, if any, will be. There is a struggle between f and g . If f wins, the limit will be 0; if g wins, the answer will be ∞ (or $-\infty$). Or there may be a compromise where the answer is a finite nonzero number. This kind of limit is called an **indeterminate form of type $0 \cdot \infty$** . We can deal with it by writing the product fg as a quotient:

$$fg = \frac{f}{1/g} \quad \text{or} \quad fg = \frac{g}{1/f}$$

This converts the given limit into an indeterminate form of type $\frac{0}{0}$ or $\frac{\infty}{\infty}$ so that we can use l'Hospital's Rule.

EXAMPLE 8 | Evaluate $\lim_{x \rightarrow 0^+} x \ln x$. Use the knowledge of this limit, together with information from derivatives, to sketch the curve $y = x \ln x$.

SOLUTION The given limit is indeterminate because, as $x \rightarrow 0^+$, the first factor (x) approaches 0 while the second factor ($\ln x$) approaches $-\infty$. Writing $x = 1/(1/x)$, we have $1/x \rightarrow \infty$ as $x \rightarrow 0^+$, so l'Hospital's Rule gives

$$\lim_{x \rightarrow 0^+} x \ln x = \lim_{x \rightarrow 0^+} \frac{\ln x}{1/x} = \lim_{x \rightarrow 0^+} \frac{1/x}{-1/x^2} = \lim_{x \rightarrow 0^+} (-x) = 0$$

If $f(x) = x \ln x$, then

$$f'(x) = x \cdot \frac{1}{x} + \ln x = 1 + \ln x$$

so $f'(x) = 0$ when $\ln x = -1$, which means that $x = e^{-1}$. In fact, $f'(x) > 0$ when $x > e^{-1}$ and $f'(x) < 0$ when $x < e^{-1}$, so f is increasing on $(1/e, \infty)$ and decreasing on $(0, 1/e)$. Thus, by the First Derivative Test, $f(1/e) = -1/e$ is a local (and absolute) minimum. Also, $f''(x) = 1/x > 0$, so f is concave upward on $(0, \infty)$. We use this information, together with the crucial knowledge that $\lim_{x \rightarrow 0^+} f(x) = 0$, to sketch the curve in Figure 7. ■

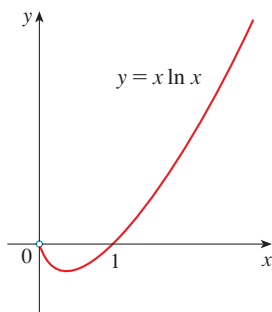


FIGURE 7

NOTE In solving Example 8 another possible option would have been to write

$$\lim_{x \rightarrow 0^+} x \ln x = \lim_{x \rightarrow 0^+} \frac{x}{1/\ln x}$$

This gives an indeterminate form of the type $\frac{0}{0}$, but if we apply l'Hospital's Rule we get a more complicated expression than the one we started with. In general, when we rewrite an indeterminate product, we try to choose the option that leads to the simpler limit.

EXAMPLE 9 | Blood alcohol concentration In Sections 3.1 and 4.1 we modeled the blood alcohol concentration (BAC) after rapid consumption of one alcoholic drink by the function

$$C(t) = 0.0225te^{-0.0467t}$$

where t is measured in minutes after consumption and C is measured in mg/mL. Calculate $\lim_{t \rightarrow \infty} C(t)$.

SOLUTION The limit is an indeterminate product because $0.0225t \rightarrow \infty$ and $e^{-0.0467t} \rightarrow 0$ as $t \rightarrow \infty$. If we write $C(t)$ as a quotient instead of a product and then use l'Hospital's Rule, we get

$$\begin{aligned} \lim_{t \rightarrow \infty} C(t) &= \lim_{t \rightarrow \infty} 0.0225te^{-0.0467t} = \lim_{t \rightarrow \infty} \frac{0.0225t}{e^{0.0467t}} \\ &= \lim_{t \rightarrow \infty} \frac{0.0225}{0.0467e^{0.0467t}} = 0 \end{aligned}$$

because the denominator approaches infinity as $t \rightarrow \infty$. This result is not surprising; it is to be expected that the BAC eventually approaches 0 as time passes. Figure 8 illustrates the result.

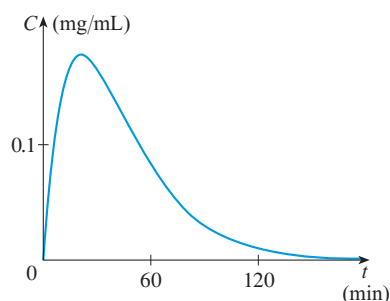


FIGURE 8

Indeterminate Differences

If $\lim_{x \rightarrow a} f(x) = \infty$ and $\lim_{x \rightarrow a} g(x) = \infty$, then the limit

$$\lim_{x \rightarrow a} [f(x) - g(x)]$$

is called an **indeterminate form of type $\infty - \infty$** . Again there is a contest between f and g . Will the answer be ∞ (f wins) or will it be $-\infty$ (g wins) or will they compromise on a finite number? To find out, we try to convert the difference into a quotient (for instance, by using a common denominator, or rationalization, or factoring out a common factor) so that we have an indeterminate form of type $\frac{0}{0}$ or $\frac{\infty}{\infty}$.

EXAMPLE 10 | Compute $\lim_{x \rightarrow (\pi/2)^-} (\sec x - \tan x)$.

SOLUTION First notice that $\sec x \rightarrow \infty$ and $\tan x \rightarrow \infty$ as $x \rightarrow (\pi/2)^-$, so the limit is indeterminate. Here we use a common denominator:

$$\begin{aligned} \lim_{x \rightarrow (\pi/2)^-} (\sec x - \tan x) &= \lim_{x \rightarrow (\pi/2)^-} \left(\frac{1}{\cos x} - \frac{\sin x}{\cos x} \right) \\ &= \lim_{x \rightarrow (\pi/2)^-} \frac{1 - \sin x}{\cos x} = \lim_{x \rightarrow (\pi/2)^-} \frac{-\cos x}{-\sin x} = 0 \end{aligned}$$

Note that the use of l'Hospital's Rule is justified because $1 - \sin x \rightarrow 0$ and $\cos x \rightarrow 0$ as $x \rightarrow (\pi/2)^-$.

EXAMPLE 11 | Calculate $\lim_{x \rightarrow \infty} (e^x - x)$.

SOLUTION This is an indeterminate difference because both e^x and x approach infinity. We would certainly expect the answer to be infinity because $e^x \rightarrow \infty$ much faster than x . But we can verify this by factoring out x :

$$e^x - x = x \left(\frac{e^x}{x} - 1 \right)$$

The term $e^x/x \rightarrow \infty$ as $x \rightarrow \infty$ by l'Hospital's Rule and so we now have a product in which both factors grow large:

$$\lim_{x \rightarrow \infty} (e^x - x) = \lim_{x \rightarrow \infty} \left[x \left(\frac{e^x}{x} - 1 \right) \right] = \infty$$

EXERCISES 4.3

1–34 Find the limit. Use l'Hospital's Rule where appropriate. If there is a more elementary method, consider using it. If l'Hospital's Rule doesn't apply, explain why.

1. $\lim_{x \rightarrow 1} \frac{x^2 - 1}{x^2 - x}$

2. $\lim_{x \rightarrow 1} \frac{x^a - 1}{x^b - 1}$

3. $\lim_{x \rightarrow (\pi/2)^+} \frac{\cos x}{1 - \sin x}$

4. $\lim_{x \rightarrow 0} \frac{\sin 4x}{\tan 5x}$

5. $\lim_{t \rightarrow 0} \frac{e^t - 1}{t^3}$

6. $\lim_{t \rightarrow 0} \frac{e^{3t} - 1}{t}$

7. $\lim_{x \rightarrow \infty} \frac{\ln x}{\sqrt{x}}$

8. $\lim_{\theta \rightarrow \pi/2} \frac{1 - \sin \theta}{\csc \theta}$

9. $\lim_{x \rightarrow 0^+} \frac{\ln x}{x}$

10. $\lim_{x \rightarrow \infty} \frac{(\ln x)^2}{x}$

11. $\lim_{x \rightarrow 0} \frac{\sqrt{1+2x} - \sqrt{1-4x}}{x}$

12. $\lim_{x \rightarrow 1} \frac{\ln x}{\sin \pi x}$

13. $\lim_{t \rightarrow 0} \frac{5^t - 3^t}{t}$

14. $\lim_{u \rightarrow \infty} \frac{e^{u/10}}{u^3}$

15. $\lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x^2}$

16. $\lim_{x \rightarrow 0} \frac{\cos mx - \cos nx}{x^2}$

17. $\lim_{x \rightarrow 1} \frac{1 - x + \ln x}{1 + \cos \pi x}$

18. $\lim_{x \rightarrow 0} \frac{x}{\tan^{-1}(4x)}$

19. $\lim_{x \rightarrow 1} \frac{x^a - ax + a - 1}{(x - 1)^2}$

20. $\lim_{x \rightarrow 0} \frac{e^x - e^{-x} - 2x}{x - \sin x}$

21. $\lim_{x \rightarrow 0} \frac{\cos x - 1 + \frac{1}{2}x^2}{x^4}$

22. $\lim_{x \rightarrow a^+} \frac{\cos x \ln(x - a)}{\ln(e^x - e^a)}$

23. $\lim_{x \rightarrow \infty} x \sin(\pi/x)$

24. $\lim_{x \rightarrow -\infty} x^2 e^x$

25. $\lim_{x \rightarrow 0} \cot 2x \sin 6x$

26. $\lim_{x \rightarrow 0^+} \sin x \ln x$

27. $\lim_{x \rightarrow \infty} x^3 e^{-x^2}$

28. $\lim_{x \rightarrow \infty} x \tan(1/x)$

29. $\lim_{x \rightarrow 1} \left(\frac{x}{x-1} - \frac{1}{\ln x} \right)$

30. $\lim_{x \rightarrow 0} (\csc x - \cot x)$

31. $\lim_{x \rightarrow \infty} (\sqrt{x^2 + x} - x)$

32. $\lim_{x \rightarrow 0^+} \left(\cot x - \frac{1}{x} \right)$

33. $\lim_{x \rightarrow \infty} (x - \ln x)$

34. $\lim_{x \rightarrow \infty} (xe^{1/x} - x)$

35–38 Use l'Hospital's Rule to help find the asymptotes of f . Then use them, together with information from f' and f'' , to sketch the graph of f . Check your work with a graphing device.

35. $f(x) = xe^{-x}$

36. $f(x) = e^x/x$

37. $f(x) = (\ln x)/x$

38. $f(x) = xe^{-x^2}$

39–42 Rank the functions in order of how quickly they grow as $x \rightarrow \infty$.

39. $y = x^5, \quad y = \ln(x^{10}), \quad y = e^{2x}, \quad y = e^{3x}$

40. $y = 2^x, \quad y = 3^x, \quad y = e^{x/2}, \quad y = e^{x/3}$

41. $y = (\ln x)^2, \quad y = (\ln x)^3, \quad y = \sqrt{x}, \quad y = \sqrt[3]{x}$

42. $y = x + e^{-x}, \quad y = 10 \ln x, \quad y = 5\sqrt{x}, \quad y = x\sqrt{x}$

43–44 Guess the value of the limit by considering the dominant terms in the numerator and denominator. Then use l'Hospital's Rule to confirm your guess.

43. $\lim_{x \rightarrow \infty} \frac{e^{-2x} + x + e^{0.1x}}{x^3 - x^2}$

44. $\lim_{x \rightarrow \infty} \frac{x^2 - x + \ln x}{x + 2^x}$

45–46 Rank the functions in order of how quickly they approach 0 as $x \rightarrow \infty$.

45. $y = \frac{1}{x}$, $y = \frac{1}{x^2}$, $y = e^{-x}$, $y = x^{-1/2}$

46. $y = e^{-x}$, $y = xe^{-x}$, $y = e^{-x^2}$, $y = xe^{-x^2}$

47–48 What happens if you try to use l'Hospital's Rule to find the limit? Evaluate the limit using another method.

47. $\lim_{x \rightarrow \infty} \frac{x}{\sqrt{x^2 + 1}}$

48. $\lim_{x \rightarrow (\pi/2)^-} \frac{\sec x}{\tan x}$

49. Models of population growth have the general form $dN/dt = f(N)$, where $f(N)$ is a function such that $f(0) = 0$ and $f(N)$ is positive for some positive values of N . The per capita growth rate is defined to be the population growth rate divided by the population size.

- What is the per capita growth rate for the model $dN/dt = rN$?
- What is the per capita growth rate for the model $dN/dt = rN(1 - N/K)$ when N is small (that is, when $N \rightarrow 0$)?
- Express the per capita growth rate when $N \rightarrow 0$ for the general model $dN/dt = f(N)$ in terms of the function f and/or its derivative.

50. Foraging In Exercise 4.2.2 we let $N(t)$ be the amount of nectar foraged from a flower by a bumblebee in t seconds.

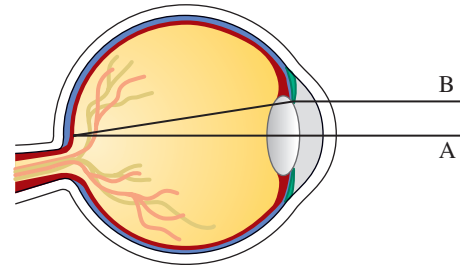
- What is the average rate of nectar consumption over a period of t seconds?
- Find the average rate of nectar consumption for very short foraging visits by using l'Hospital's Rule to calculate the limit of the answer to part (a) as $t \rightarrow 0$.

51. Stiles-Crawford effect Light enters the eye through the pupil and strikes the retina, where photoreceptor cells sense light and color. W. Stanley Stiles and B. H. Crawford studied the phenomenon in which measured brightness decreases as light enters farther from the center of the pupil. (See the figure.) They detailed their findings of this phenomenon, known as the *Stiles-Crawford effect of the first kind*, in a paper published in 1933. In particular, they observed that the amount of luminance sensed was not proportional to the area of the pupil. The percentage P of the luminance entering a pupil at a distance of r mm from the center that is

sensed at the retina can be described by

$$P = \frac{100(1 - 10^{-\rho r^2})}{\rho r^2 \ln 10}$$

where ρ is an experimentally determined constant, typically about 0.05.



A light beam A that enters through the center of the pupil measures brighter than a beam B entering near the edge of the pupil.

- If light enters a pupil at a distance of 3 mm from the center, what is the percentage of luminance that is sensed at the retina?
- Compute the percentage of luminance sensed if light enters the pupil at a distance of 2 mm from the center. Does it make sense that it is larger than the answer to part (a)?
- Compute $\lim_{r \rightarrow 0} P$. Is the result what you would expect?

Source: Adapted from W. Stiles et al., "The Luminous Efficiency of Rays Entering the Eye Pupil at Different Points." *Proceedings of the Royal Society of London. Series B*, 112 (1933): 428–50.

52. Antibiotic concentration for large patients Suppose an antibiotic is administered at a constant rate through intravenous supply to a patient and is metabolized. It can be shown using the type of mixing models discussed in Exercises 7.4.45–48 that the concentration of antibiotic after one unit of time is

$$c(V) = c_0 e^{-1/V} + \theta V(1 - e^{-1/V})$$

where c_0 is the initial concentration, θ is the rate of supply, and V is the volume of the patient's blood. What is the predicted value for large patients (that is, for large values of V)?

53. Drug pharmacokinetics So-called two-compartment models are often used to describe drug pharmacokinetics, with the blood being one compartment and the internal organs being the other (see Section 10.3). If the rate of flow from the blood to the organs is α and the rate of metabolism from the organs is β , then under certain conditions the concentration of drug in the organ at time t is given by

$$\frac{e^{-\alpha t} - e^{-\beta t}}{\beta - \alpha}$$

What is the predicted concentration at time t if the values of α and β are very close to one another?

- 54. Drug pharmacokinetics** The level of medication in the bloodstream after a drug is administered is often modeled by a function of the form

$$S(t) = At^p e^{-kt}$$

where A , p , and k are positive constants. This is called a *surge function* because its graph shows an initial surge in the drug level followed by a more gradual decline. (Particular cases have been investigated in Example 9 and Exercise 4.2.51.)

- (a) Use l'Hospital's Rule to show that $\lim_{t \rightarrow \infty} S(t) = 0$ for all positive values of A , p , and k .
- (b) Investigate the family of surge functions for $A = 1$ and positive values of p and k . What features do these curves have in common? How do they differ from one another? In particular, what happens to the maximum and minimum points and inflection points as p and k change? Illustrate by graphing several members of the family.

- 55.** Prove that

$$\lim_{x \rightarrow \infty} \frac{e^x}{x^n} = \infty$$

for any positive integer n . This shows that the exponential function approaches infinity faster than any power of x .

- 56.** Prove that

$$\lim_{x \rightarrow \infty} \frac{\ln x}{x^p} = 0$$

for any number $p > 0$. This shows that the logarithmic function approaches infinity more slowly than any power of x .

- 57.** The first appearance in print of l'Hospital's Rule was in the book *Analyse des Infiniment Petits* published by the Marquis de l'Hospital in 1696. This was the first calculus textbook ever published and the example that the Marquis used in that book to illustrate his rule was to find the limit of the function

$$y = \frac{\sqrt{2a^3x - x^4} - a\sqrt[3]{aax}}{a - \sqrt[4]{ax^3}}$$

as x approaches a , where $a > 0$. (At that time it was common to write aa instead of a^2 .) Solve this problem.

- 58.** If f' is continuous, $f(2) = 0$, and $f'(2) = 7$, evaluate

$$\lim_{x \rightarrow 0} \frac{f(2 + 3x) + f(2 + 5x)}{x}$$

PROJECT Mutation-Selection Balance in Genetic Diseases

Several human diseases—such as Tay-Sachs disease, phenylketonuria, neurofibromatosis, and Huntington's disease—occur if an individual carries a mutated copy of a specific gene. Because the carriers of such deleterious genes are less likely to survive and reproduce, the frequency of such genes in the population is usually quite low.

In population-genetic terms the frequency of such disease-causing genes reflects a balance between their spontaneous appearance through mutation and selection against them through reduced survival. Moreover, some diseases (like Tay-Sachs and phenylketonuria) are *recessive*, meaning that only individuals carrying two copies of the mutated gene are affected, while others (like neurofibromatosis and Huntington's disease) are *dominant*, meaning that individuals carrying even a single copy will be affected.

Let's model the mutation-selection balance for an arbitrary disease to predict the frequency of the mutation in the population. We will use **A** to denote the disease-causing variant and **a** to denote the normal version of the gene. For simplicity we will assume that generations are discrete in time and we will model the dynamics of the disease-causing gene using a difference equation.

Suppose the probability of surviving one time step for an **aa** individual is standardized to 1, and that for an **AA** individual is $1 - s$, where s is a constant and $0 < s < 1$. We can incorporate different levels of dominance by supposing that the probability of survival of **Aa** individuals is $1 - sh$, where h is a constant and $0 \leq h \leq 1$. When $h = 1$ the disease is dominant since **Aa** individuals are just as affected by the disease as **AA** individuals. When $h = 0$ the disease is recessive because only **AA** individuals are affected. The case $h = \frac{1}{2}$ is called *codominant*, meaning that the survival of **Aa** individuals lies exactly halfway between those of **AA** and **aa** individuals.

To describe how selection will change the frequency of the **A** gene, we can follow a derivation similar to that used in the project on page 78. Using p_t for the frequency of **A** at time t , we obtain the following expression for the frequency of **A** after selection has occurred:

$$\frac{p_t^2(1-s) + p_t(1-p_t)(1-sh)}{p_t^2(1-s) + 2p_t(1-p_t)(1-sh) + (1-p_t)^2}$$

1. Suppose that, in a single time step, first selection occurs as described by this expression and then a fraction μ of the **a** variants mutate to the disease-causing variant **A**. Obtain an expression for p_{t+1} in terms of p_t .
2. At mutation-selection balance, the frequency of **A** will no longer change, meaning that $p_{t+1} = p_t$. Using $\hat{p}$ to denote this equilibrium frequency, show that $\hat{p}$ satisfies a cubic equation. Further, show that this cubic equation can be reduced to a quadratic equation if we are interested only in solutions satisfying $\hat{p} \neq 1$.
3. Show that one root of the quadratic equation obtained in Problem 2 is

$$\hat{p} = \frac{sh(1+\mu) - \sqrt{[sh(1+\mu)]^2 - 4s\mu(2h-1)}}{2s(2h-1)}$$

4. Assume that $2\mu < s$. What is the predicted frequency of genes that cause recessive diseases (that is, the frequency as $h \rightarrow 0$)?
5. Assume that $2\mu < s$. Use l'Hospital's Rule to determine the predicted frequency of genes that cause codominant diseases (that is, the frequency as $h \rightarrow \frac{1}{2}$).

4.4 Optimization Problems

There are many situations in the life sciences in which it is desirable to find an optimal outcome: maximizing the yield of an agricultural crop by controlling the nitrogen level of the soil, minimizing the energy required for fish to swim or birds to fly during migration, determining the dosage of a drug for the best result. In this section we use the methods of this chapter to determine such optimal outcomes.

In solving these problems the greatest challenge is often to convert the word problem into a mathematical optimization problem by setting up the function that is to be maximized or minimized. The following steps may be useful.

Steps in Solving Optimization Problems

1. **Understand the Problem** The first step is to read the problem carefully until it is clearly understood. Ask yourself: What is the unknown? What are the given quantities? What are the given conditions?
2. **Draw a Diagram** In most problems it is useful to draw a diagram and identify the given and required quantities on the diagram.
3. **Introduce Notation** Assign a symbol to the quantity that is to be maximized or minimized (let's call it Q for now). Also select symbols ($a, b, c, \dots, x, y$) for other unknown quantities and label the diagram with these symbols. It may help to use initials as suggestive symbols—for example, A for area, h for height, t for time.

4. Express Q in terms of some of the other symbols from Step 3.
5. If Q has been expressed as a function of more than one variable in Step 4, use the given information to find relationships (in the form of equations) among these variables. Then use these equations to eliminate all but one of the variables in the expression for Q . Thus Q will be expressed as a function of *one* variable x , say, $Q = f(x)$. Write the domain of this function.
6. Use the methods of Sections 4.1 and 4.2 to find the *absolute* maximum or minimum value of f . In particular, if the domain of f is a closed interval, then the Closed Interval Method in Section 4.1 can be used.

EXAMPLE 1 | A farmer has 2400 ft of fencing and wants to fence off a rectangular field that borders a river with a straight bank. He needs no fence along the river. What are the dimensions of the field that has the largest area?

SOLUTION In order to get a feeling for what is happening in this problem, let's experiment with some special cases. Figure 1 (not to scale) shows three possible ways of laying out the 2400 ft of fencing.

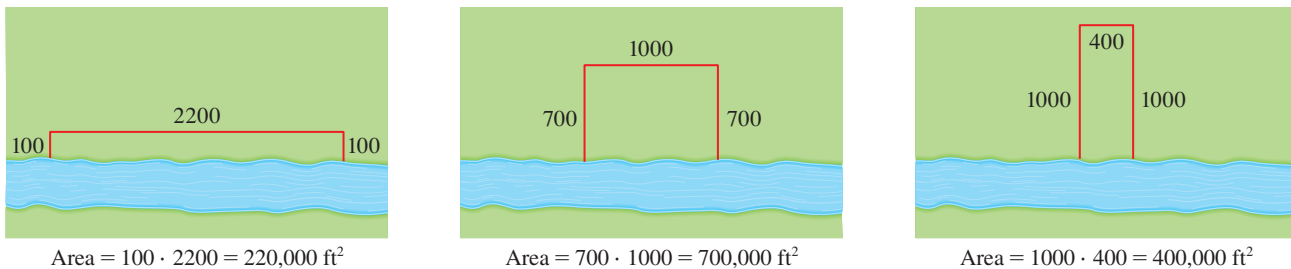


FIGURE 1

We see that when we try shallow, wide fields or deep, narrow fields, we get relatively small areas. It seems plausible that there is some intermediate configuration that produces the largest area.

Figure 2 illustrates the general case. We wish to maximize the area A of the rectangle. Let x and y be the depth and width of the rectangle (in feet). Then we express A in terms of x and y :

$$A = xy$$

We want to express A as a function of just one variable, so we eliminate y by expressing it in terms of x . To do this we use the given information that the total length of the fencing is 2400 ft. Thus

$$2x + y = 2400$$

From this equation we have $y = 2400 - 2x$, which gives

$$A = x(2400 - 2x) = 2400x - 2x^2$$

Note that $x \geq 0$ and $x \leq 1200$ (otherwise $A < 0$). So the function that we wish to maximize is

$$A(x) = 2400x - 2x^2 \quad 0 \leq x \leq 1200$$

The derivative is $A'(x) = 2400 - 4x$, so to find the critical numbers we solve the

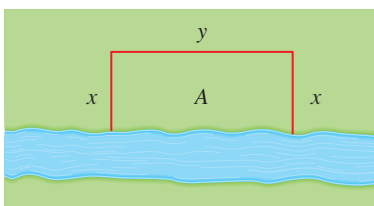


FIGURE 2

equation

$$2400 - 4x = 0$$

which gives $x = 600$. The maximum value of A must occur either at this critical number or at an endpoint of the interval. Since $A(0) = 0$, $A(600) = 720,000$, and $A(1200) = 0$, the Closed Interval Method gives the maximum value as $A(600) = 720,000$.

[Alternatively, we could have observed that $A''(x) = -4 < 0$ for all x , so A is always concave downward and the local maximum at $x = 600$ must be an absolute maximum.]

Thus the rectangular field should be 600 ft deep and 1200 ft wide. ■



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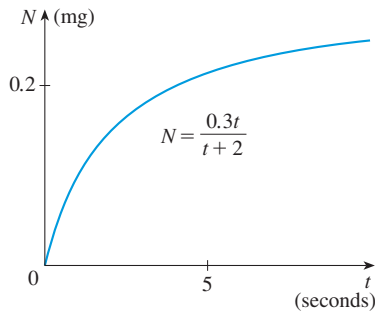


FIGURE 3

EXAMPLE 2 | **BB** Nectar foraging by bumblebees Many animals forage on resources that are distributed in discrete patches. For example, bumblebees visit many flowers, foraging on nectar from each. The amount of nectar $N(t)$ consumed from any flower increases with the amount of time spent at that flower, but with diminishing returns (see Figure 3). Suppose this function is given by

$$N(t) = \frac{0.3t}{t + 2}$$

where t is measured in seconds and N in milligrams. Suppose also that the time it takes a bee to travel from one flower to the next is 4 seconds.

- (a) If a bee spends t seconds at each flower, find an equation for the average amount of nectar consumed per second, from the beginning of a visit to a flower until the beginning of the visit to the next flower.
- (b) Suppose bumblebees forage on a given flower for an amount of time that maximizes the average rate of energy gain obtained in part (a). What is this optimal foraging time?

SOLUTION

- (a) The total time to complete one cycle of foraging and traveling is $t + 4$. So the average amount of nectar consumed per second over one cycle is

$$f(t) = \frac{N(t)}{t + 4} = \frac{0.3t}{(t + 2)(t + 4)} = \frac{0.3t}{t^2 + 6t + 8}$$

- (b) The derivative of f is

$$f'(t) = \frac{(t^2 + 6t + 8)(0.3) - (0.3t)(2t + 6)}{(t^2 + 6t + 8)^2} = \frac{-0.3t^2 + 2.4}{(t^2 + 6t + 8)^2}$$

Then $f'(t) = 0$ when $t^2 = 2.4/0.3 = 8$. But $t \geq 0$, so the only critical number is $t = \sqrt{8} = 2\sqrt{2}$.

Since the domain of f is $[0, \infty)$, we can't use the argument of Example 1 concerning endpoints. But we observe that $f'(t) > 0$ for $0 \leq t < 2\sqrt{2}$ and $f'(t) < 0$ for $t > 2\sqrt{2}$, so f is increasing for all t to the left of the critical number and decreasing for all t to the right. So the optimal foraging time is

$$t = 2\sqrt{2} \approx 2.83 \text{ seconds}$$

This result is illustrated by the graph of f in Figure 4. ■

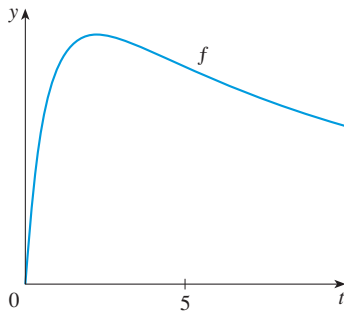


FIGURE 4

NOTE 1 The argument used in Example 2 to justify the absolute maximum is a variant of the First Derivative Test (which applies only to *local* maximum or minimum values) and is stated here for future reference.

TEC Module 4.4 takes you through six additional optimization problems, including animations of the physical situations.

First Derivative Test for Absolute Extreme Values Suppose that c is a critical number of a continuous function f defined on an interval.

- (a) If $f'(x) > 0$ for all $x < c$ and $f'(x) < 0$ for all $x > c$, then $f(c)$ is the absolute maximum value of f .
- (b) If $f'(x) < 0$ for all $x < c$ and $f'(x) > 0$ for all $x > c$, then $f(c)$ is the absolute minimum value of f .

NOTE 2 In Example 4.1.7 we found the maximum value of the blood alcohol concentration during the first hour by using the Closed Interval Method. But if we use the First Derivative Test for Absolute Extreme Values, we can see that it is the absolute maximum. The formula

$$C'(t) = 0.0225e^{-0.0467t}(1 - 0.0467t)$$

shows that $C'(t) > 0$ for all $t < 1/0.0467$ and $C'(t) < 0$ for all $t > 1/0.0467$. It follows that the absolute maximum of the BAC is $C(1/0.0467) \approx 0.177$ mg/mL.

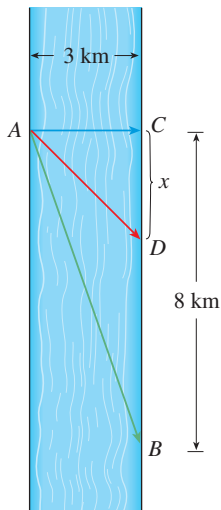


FIGURE 5

EXAMPLE 3 | A man launches his boat from point A on a bank of a straight water channel, 3 km wide, and wants to reach point B, 8 km south on the opposite bank, as quickly as possible (see Figure 5). He could row his boat directly across the channel to point C and then run to B, or he could row directly to B, or he could row to some point D between C and B and then run to B. If he can row 6 km/h and run 8 km/h, where should he land to reach B as soon as possible? (We assume that the water in the channel is not moving.)

SOLUTION If we let x be the distance from C to D, then the running distance is $|DB| = 8 - x$ and the Pythagorean Theorem gives the rowing distance as $|AD| = \sqrt{x^2 + 9}$. We use the equation

$$\text{time} = \frac{\text{distance}}{\text{rate}}$$

Then the rowing time is $\sqrt{x^2 + 9}/6$ and the running time is $(8 - x)/8$, so the total time T as a function of x is

$$T(x) = \frac{\sqrt{x^2 + 9}}{6} + \frac{8 - x}{8}$$

The domain of this function T is $[0, 8]$. Notice that if $x = 0$, he rows to C and if $x = 8$, he rows directly to B. The derivative of T is

$$T'(x) = \frac{x}{6\sqrt{x^2 + 9}} - \frac{1}{8}$$

Thus, using the fact that $x \geq 0$, we have

$$\begin{aligned} T'(x) = 0 &\iff \frac{x}{6\sqrt{x^2 + 9}} = \frac{1}{8} &\iff 4x = 3\sqrt{x^2 + 9} \\ &\iff 16x^2 = 9(x^2 + 9) &\iff 7x^2 = 81 \\ &\iff x = \frac{9}{\sqrt{7}} \end{aligned}$$

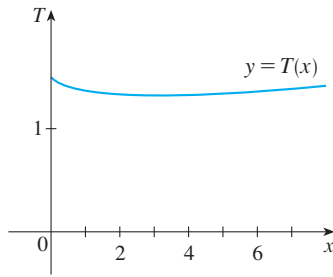


FIGURE 6

The only critical number is $x = 9/\sqrt{7}$. To see whether the minimum occurs at this critical number or at an endpoint of the domain $[0, 8]$, we evaluate T at all three points:

$$T(0) = 1.5 \quad T\left(\frac{9}{\sqrt{7}}\right) = 1 + \frac{\sqrt{7}}{8} \approx 1.33 \quad T(8) = \frac{\sqrt{73}}{6} \approx 1.42$$

Since the smallest of these values of T occurs when $x = 9/\sqrt{7}$, the absolute minimum value of T must occur there. Figure 6 illustrates this calculation by showing the graph of T .

Thus the man should land the boat at a point $9/\sqrt{7}$ km (≈ 3.4 km) south from his starting point. ■



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EXAMPLE 4 | **BB** Aquatic birds¹ forage underwater and periodically return to the surface to replenish their oxygen stores. Oxygen stores increase with the amount of time spent on the surface but in a diminishing way, according to the model

$$O(t) = \frac{20t}{5 + t}$$

where t is the amount of time spent at the surface (in seconds). Suppose the round-trip travel time to and from the underwater foraging area is T seconds and oxygen is depleted at a constant rate of r mL/s while a bird is underwater. Furthermore, suppose the bird forages until it has just enough oxygen to return to the surface.

- If Q is the fraction of a single dive cycle that the bird spends foraging, find an equation for Q as a function of the surface time t .
- If $T = 2$ seconds and $r = 1$ mL/s, find the surface time t that maximizes the fraction of time spent foraging.

SOLUTION

- Let f be the time spent foraging during a single cycle. Because the surface time is t seconds and the total travel time for a single cycle is T seconds, the fraction of a cycle spent foraging is

$$Q = \frac{f}{t + T + f}$$

This expression for Q involves the constant T and the two variables t and f . We want to express Q in terms of just one variable and so we eliminate f by expressing it in terms

1. L. Halsey et al., “Optimal Diving Behaviour and Respiratory Gas Exchange in Birds,” *Respiratory Physiology and Neurobiology* 154 (2006): 268–83.

of t . To do this we use the information given—that all of the oxygen $O(t)$ obtained at the surface is used for foraging and travel. In other words, $O(t) = rT + rf$. Solving this equation for f gives

$$f = \frac{O(t) - rT}{r}$$

Therefore

$$\begin{aligned} Q(t) &= \frac{\frac{O(t) - rT}{r}}{t + T + \frac{O(t) - rT}{r}} = \frac{O(t) - rT}{O(t) + rT} \\ &= \frac{\frac{20t}{5+t} - rT}{\frac{20t}{5+t} + rT} = \frac{(20 - rT)t - 5rT}{20t + 5rt + rt^2} \end{aligned}$$

(b) With $r = 1$ and $T = 2$ we have

$$Q(t) = \frac{18t - 10}{t^2 + 25t}$$

and so the derivative of Q is

$$Q'(t) = \frac{(t^2 + 25t)(18) - (18t - 10)(2t + 25)}{(t^2 + 25t)^2} = -\frac{2(9t^2 - 10t - 125)}{(t^2 + 25t)^2}$$

Solving the equation $Q'(t) = 0$ with the quadratic formula, and retaining only the positive root, we get

$$t = \frac{5}{9}(1 + \sqrt{46}) \approx 4.32 \text{ seconds}$$

This value of t gives the largest fraction of foraging time because $Q'(t)$ is positive to the left and negative to the right of $t = \frac{5}{9}(1 + \sqrt{46})$. ■

EXAMPLE 5 | **BB** Sustainable harvesting² For many natural fish populations, the net number of new recruits to the population in a given year can be modeled as a function of the existing population size N by an equation of the form

$$R(N) = rN \left(1 - \frac{N}{K} \right)$$

where r and K are positive constants. (K is called the carrying capacity.) The population will increase if the net number of recruits $R(N)$ is positive and it will decrease if $R(N)$ is negative. Thus, because $R(N)$ is positive when $0 < N < K$ and $R(N)$ is negative when $N > K$, we expect the population to stabilize at a constant size of $N = K$.

If the population is subject to harvesting, N will begin to change and once the population has stabilized again, the number of fish harvested each year, which we denote by H , must equal the net recruitment for that year; that is, $R(N) = H$.

(a) Suppose $H = hN$, where h is a measure of the “fishing effort” expended. What is the population size once it has stabilized?

2. C. Clark, “The Economics of Overexploitation,” *Science* 181 (1973): 630–34.

- (b) Express the total harvest rate H as a function of h once the population has stabilized, and determine the fishing effort that results in the largest possible harvest rate.
- (c) What is the population size once it has stabilized if this fishing effort is used, and what is the total harvest?

SOLUTION

- (a) Since $H = hN$, once the population has stabilized we have

$$hN = H = rN \left(1 - \frac{N}{K} \right)$$

and so

$$h = r \left(1 - \frac{N}{K} \right) \quad \Rightarrow \quad \frac{rN}{K} = r - h$$

Solving this equation for N , we get

$$N = \frac{K}{r}(r - h) = K \left(1 - \frac{h}{r} \right)$$

This is the population size once it has become stable.

- (b) With the expression for N from part (a), we have

$$H = hN = K \left(h - \frac{h^2}{r} \right)$$

and so
$$\frac{dH}{dh} = K \left(1 - \frac{2h}{r} \right) = 0 \quad \text{when } h = \frac{r}{2}$$

Also, dH/dh is positive when $h < \frac{1}{2}r$ and negative when $h > \frac{1}{2}r$. So the largest possible total harvest rate occurs when $h = \frac{1}{2}r$.

- (c) If this optimal fishing effort is used, then the ultimate population size is

$$N = K \left(1 - \frac{\frac{1}{2}r}{r} \right) = K \left(1 - \frac{1}{2} \right) = \frac{K}{2}$$

and the total harvest is

$$H = hN = \frac{r}{2} \cdot \frac{K}{2} = \frac{rK}{4}$$

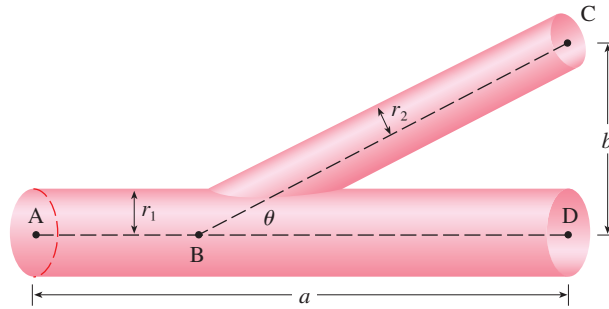
EXAMPLE 6 | Branching blood vessels The blood vascular system consists of blood vessels (arteries, arterioles, capillaries, and veins) that convey blood from the heart to the organs and back to the heart. This system should work so as to minimize the energy expended by the heart in pumping the blood. In particular, this energy is reduced when the resistance of the blood is lowered. One of Poiseuille's Laws gives the resistance R of the blood as

$$R = C \frac{L}{r^4}$$

where L is the length of the blood vessel, r is the radius, and C is a positive constant determined by the viscosity of the blood. (Poiseuille established this law experimen-

tally, but it also follows from the results of Section 6.3.) Figure 7 shows a main blood vessel with radius r_1 branching at an angle θ into a smaller vessel with radius r_2 .

FIGURE 7
Vascular branching



(a) Express the total resistance R of the blood along the path ABC as a function of the branching angle θ .

(b) Show that R is minimized when

$$\cos \theta = \frac{r_2^4}{r_1^4}$$

(c) Find the optimal branching angle (correct to the nearest degree) when the radius of the smaller blood vessel is two-thirds the radius of the larger vessel.

SOLUTION

(a) From Figure 7 we see that

$$\sin \theta = \frac{b}{|BC|} \quad \Rightarrow \quad |BC| = b \csc \theta$$

and

$$\cos \theta = \frac{|BD|}{|BC|} \quad \Rightarrow \quad |BC| = (a - |AB|) \sec \theta$$

Equating the expressions for $|BC|$, we have

$$b \csc \theta = (a - |AB|) \sec \theta \quad \Rightarrow \quad b \cot \theta = a - |AB|$$

$$\text{so} \quad |AB| = a - b \cot \theta$$

Then, from Poiseuille's Law, the total resistance is

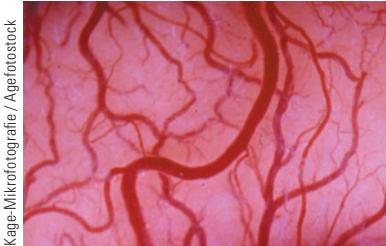
$$R(\theta) = C \frac{|AB|}{r_1^4} + C \frac{|BC|}{r_2^4} = C \left(\frac{a - b \cot \theta}{r_1^4} + \frac{b \csc \theta}{r_2^4} \right)$$

(b) Differentiating the expression for $R(\theta)$ in part (a), we get

$$R'(\theta) = C \left(\frac{b \csc^2 \theta}{r_1^4} - \frac{b \csc \theta \cot \theta}{r_2^4} \right) = bC \csc \theta \left(\frac{\csc \theta}{r_1^4} - \frac{\cot \theta}{r_2^4} \right)$$

So

$$R'(\theta) = 0 \quad \Leftrightarrow \quad \frac{\csc \theta}{r_1^4} = \frac{\cot \theta}{r_2^4} \quad \Leftrightarrow \quad \frac{r_2^4}{r_1^4} = \frac{\cot \theta}{\csc \theta} = \cos \theta$$



Kage-Mikrofotografie / Agefotostock

Now

$$R'(\theta) > 0 \iff \frac{\csc \theta}{r_1^4} > \frac{\cot \theta}{r_2^4} \iff \cos \theta < \frac{r_2^4}{r_1^4}$$

and

$$R'(\theta) < 0 \quad \text{when} \quad \cos \theta > \frac{r_2^4}{r_1^4}$$

Noting that $\cos \theta$ is a decreasing function of θ for $0 < \theta < \pi$, we conclude from the First Derivative Test that the resistance has an absolute minimum value when

$$\cos \theta = \frac{r_2^4}{r_1^4}$$

(c) When $r_2 = \frac{2}{3}r_1$ we have $\cos \theta = (\frac{2}{3})^4$ and so

$$\theta = \cos^{-1}(\frac{2}{3})^4 \approx 79^\circ$$

EXERCISES 4.4

1. Consider the following problem: Find two numbers whose sum is 23 and whose product is a maximum.
- (a) Make a table of values, like the following one, so that the sum of the numbers in the first two columns is always 23. On the basis of the evidence in your table, estimate the answer to the problem.

First number	Second number	Product
1	22	22
2	21	42
3	20	60
$\vdots$	$\vdots$	$\vdots$

- (b) Use calculus to solve the problem and compare with your answer to part (a).
2. Find two numbers whose difference is 100 and whose product is a minimum.
3. Find two positive numbers whose product is 100 and whose sum is a minimum.
4. The sum of two positive numbers is 16. What is the smallest possible value of the sum of their squares?
5. Find the dimensions of a rectangle with perimeter 100 m whose area is as large as possible.
6. **Photosynthesis** The rate (in mg carbon/m³/h) at which photosynthesis takes place for a species of phytoplankton is modeled by the function

$$P = \frac{100I}{I^2 + I + 4}$$

where I is the light intensity (measured in thousands of foot-candles). For what light intensity is P a maximum?

7. **Crop yield** A model used for the yield Y of an agricultural crop as a function of the nitrogen level N in the soil (measured in appropriate units) is

$$Y = \frac{kN}{1 + N^2}$$

where k is a positive constant. What nitrogen level gives the best yield?

8. The **measles pathogenesis** function

$$f(t) = -t(t - 21)(t + 1)$$

is used in Section 5.1 to model the development of the disease, where t is measured in days and $f(t)$ represents the number of infected cells per milliliter of plasma. What is the peak infection time for the measles virus?

9. Consider the following problem: A farmer with 750 ft of fencing wants to enclose a rectangular area and then divide it into four pens with fencing parallel to one side of the rectangle. What is the largest possible total area of the four pens?
- (a) Draw several diagrams illustrating the situation, some with shallow, wide pens and some with deep, narrow pens. Find the total areas of these configurations. Does it appear that there is a maximum area? If so, estimate it.
- (b) Draw a diagram illustrating the general situation. Introduce notation and label the diagram with your symbols.
- (c) Write an expression for the total area.
- (d) Use the given information to write an equation that relates the variables.

- (e) Use part (d) to write the total area as a function of one variable.
- (f) Finish solving the problem and compare the answer with your estimate in part (a).
10. Consider the following problem: A box with an open top is to be constructed from a square piece of cardboard, 3 ft wide, by cutting out a square from each of the four corners and bending up the sides. Find the largest volume that such a box can have.
- (a) Draw several diagrams to illustrate the situation, some short boxes with large bases and some tall boxes with small bases. Find the volumes of several such boxes. Does it appear that there is a maximum volume? If so, estimate it.
- (b) Draw a diagram illustrating the general situation. Introduce notation and label the diagram with your symbols.
- (c) Write an expression for the volume.
- (d) Use the given information to write an equation that relates the variables.
- (e) Use part (d) to write the volume as a function of one variable.
- (f) Finish solving the problem and compare the answer with your estimate in part (a).
11. If 1200 cm² of material is available to make a box with a square base and an open top, find the largest possible volume of the box.
12. A box with a square base and open top must have a volume of 32,000 cm³. Find the dimensions of the box that minimize the amount of material used.
13. (a) Show that of all the rectangles with a given area, the one with smallest perimeter is a square.
- (b) Show that of all the rectangles with a given perimeter, the one with greatest area is a square.
14. A rectangular storage container with an open top is to have a volume of 10 m³. The length of its base is twice the width. Material for the base costs \$10 per square meter. Material for the sides costs \$6 per square meter. Find the cost of materials for the cheapest such container.
15. Find the point on the line $y = 2x + 3$ that is closest to the origin.
16. Find the point on the curve $y = \sqrt{x}$ that is closest to the point (3, 0).
17. **Age and size at maturity** Most organisms grow for a period of time before maturing reproductively. For many species of insects and fish, the later the age a at maturity, the larger the individual will be, and this translates into a greater reproductive output. At the same time, however, the probability of surviving to maturity decreases as the age of maturity increases. These contrasting effects can be combined into a single measure of reproductive success in

different ways. Suppose μ is a constant representing mortality rate. Find the optimal age at maturity for the following models.

$$(a) r = \frac{\ln(ae^{-\mu a})}{a} \quad (b) R = ae^{-\mu a}$$

Source: Adapted from D. Roff, *The Evolution of Life Histories: Theory and Analysis* (New York: Chapman and Hall, 1992).

18. **Enzootic stability** Suppose the rate at which people of age a get infected with a pathogen is given by $\lambda e^{-\lambda a}$, where λ is a positive constant. Not all infections develop into disease. Suppose that, of those individuals of age a that get infected, a fraction pa develop the disease (where p is a number chosen so that $0 < pa < 1$ over the ages of interest). At what age is the rate of disease development the highest?
- Source: Adapted from P. Coleman et al., "Endemic Stability—A Veterinary Idea Applied to Public Health," *The Lancet* 357 (2001): 1284–86.
19. If $C(x)$ is the cost of producing x units of a commodity, then the **average cost** per unit is $c(x) = C(x)/x$. The **marginal cost** is the rate of change of the cost with respect to the number of items produced, that is, the derivative $C'(x)$.
- (a) Show that if the average cost is a minimum, then the marginal cost equals the average cost.
- (b) If $C(x) = 16,000 + 200x + 4x^{3/2}$, in dollars, find
- (i) the cost, average cost, and marginal cost at a production level of 1000 units; (ii) the production level that will minimize the average cost; and (iii) the minimum average cost.
20. If $R(x)$ is the revenue that a company receives when it sells x units of a product, then the **marginal revenue function** is the derivative $R'(x)$. The profit function is

$$P(x) = R(x) - C(x)$$

where C is the cost function from Exercise 19.

- (a) Show that if the profit $P(x)$ is a maximum, then the marginal revenue equals the marginal cost.
- (b) If $C(x) = 16,000 + 500x - 1.6x^2 + 0.004x^3$ is the cost function and $R(x) = 1700x - 7x^2$ is the revenue function, find the production level that will maximize profit.
21. **Sustainable harvesting** Example 5 was based on the assumption that we want to maximize the total harvest size H , but instead we might want to maximize profit.
- (a) Suppose the selling price of a unit of harvest is p dollars and the cost per unit harvested is C dollars. (Assume $p > C$.) Show that the fishing effort that maximizes profit is the same as the effort that maximizes the harvest size H .
- (b) Suppose the selling price of a unit of harvest is p dollars and the unit cost is inversely proportional to the fishing effort h (that is, $C = \alpha/h$). Assume $\alpha < rp$. What is the fishing effort that maximizes profit? What is the limiting population size, assuming this fishing effort is used?

- (c) Explain how and why your answer to part (b) differs from that of part (a).

22. The **von Bertalanffy model** for the growth of an individual fish assumes that an individual acquires energy at a rate that is proportional to its length squared, but that it uses energy for metabolism at a rate proportional to its length cubed. The underlying idea is that energy acquisition is proportional to a fish's surface area, while energy utilization for metabolism is proportional to its mass (and thus its volume). Suppose the net energy available for growth at any time is the difference between acquisition and utilization through metabolism. What is the length at which growth will be fastest, as a function of the constants of proportionality?

23. **Nectar foraging by bumblebees** Suppose that, instead of the specific nectar function in Example 2, we have an arbitrary function N with $N(0) = 0$, $N(t) \geq 0$, $N'(t) > 0$, $N''(t) < 0$, and arbitrary travel time T .
- (a) Interpret the conditions on the function N .
- (b) Show that the optimal foraging time t satisfies the equation

$$N'(t) = \frac{N(t)}{t + T}$$

- (c) Show that, for any foraging time t satisfying the equation in part (b), the second derivative condition for a maximum value of the foraging function f in Example 2 is satisfied.

24. **Aquatic birds** Suppose that, instead of the specific oxygen function in Example 4, we have an arbitrary function O with $O(0) = 0$, $O(t) \geq 0$, $O'(t) > 0$, $O''(t) < 0$, and arbitrary travel time T .
- (a) Interpret the conditions on the function O .
- (b) Show that the optimal surface time t satisfies the equation

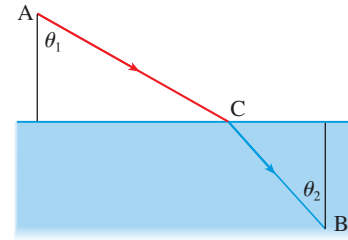
$$O'(t) = \frac{O(t) - rT}{t + T}$$

- (c) Show that, for any surface time t satisfying the equation in part (b), the second derivative condition for a maximum value of the foraging function Q in Example 4 is satisfied.

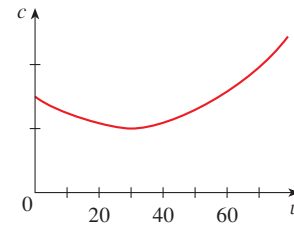
25. Let v_1 be the velocity of light in air and v_2 the velocity of light in water. According to Fermat's Principle, a ray of light will travel from a point A in the air to a point B in the water by a path ACB that minimizes the time taken. Show that

$$\frac{\sin \theta_1}{\sin \theta_2} = \frac{v_1}{v_2}$$

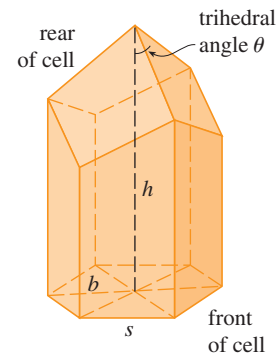
where θ_1 (the angle of incidence) and θ_2 (the angle of refraction) are as shown. This equation is known as Snell's Law.



26. The graph shows the fuel consumption c of a car (measured in gallons per hour) as a function of the speed v of the car. At very low speeds the engine runs inefficiently, so initially c decreases as the speed increases. But at high speeds the fuel consumption increases. You can see that $c(v)$ is minimized for this car when $v \approx 30$ mi/h. However, for fuel efficiency, what must be minimized is not the consumption in gallons per hour but rather the fuel consumption in gallons *per mile*. Let's call this consumption G . Using the graph, estimate the speed at which G has its minimum value.



27. **Beehives** In a beehive, each cell is a regular hexagonal prism, open at one end with a trihedral angle at the other end as in the figure.



It is believed that bees form their cells in such a way as to minimize the surface area for a given side length and height, thus using the least amount of wax in cell construction. Examination of these cells has shown that the measure

of the apex angle θ is amazingly consistent. Based on the geometry of the cell, it can be shown that the surface area S is given by

$$S = 6sh - \frac{3}{2}s^2 \cot \theta + (3s^2\sqrt{3}/2) \csc \theta$$

where s , the length of the sides of the hexagon, and h , the height, are constants.

- Calculate $dS/d\theta$.
- What angle should the bees prefer?
- Determine the minimum surface area of the cell (in terms of s and h).

Note: Actual measurements of the angle θ in beehives have been made, and the measures of these angles seldom differ from the calculated value by more than 2° .

- 28. Swimming speed of fish** For a fish swimming at a speed v relative to the water, the energy expenditure per unit time is proportional to v^3 . It is believed that migrating fish try to minimize the total energy required to swim a fixed distance. If the fish are swimming against a current u ($u < v$), then the time required to swim a distance L is $L/(v - u)$ and the total energy E required to swim the distance is given by

$$E(v) = av^3 \cdot \frac{L}{v - u}$$

where a is the proportionality constant.

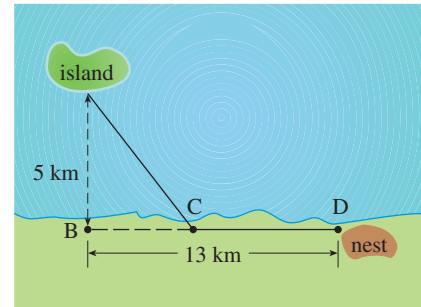
- Determine the value of v that minimizes E .
- Sketch the graph of E .

Note: This result has been verified experimentally; migrating fish swim against a current at a speed 50% greater than the current speed.

- 29. Bird flight paths** Ornithologists have determined that some species of birds tend to avoid flights over large bodies of water during daylight hours. It is believed that more energy is required to fly over water than over land because air generally rises over land and falls over water during the day. A bird with these tendencies is released from an island that is 5 km from the nearest point B on a straight shoreline, flies to a point C on the shoreline, and then flies along the shoreline to its nesting area D. Assume that the bird instinctively chooses a path that will minimize its energy expenditure. Points B and D are 13 km apart.
- In general, if it takes 1.4 times as much energy to fly over water as it does over land, to what point C should the bird fly in order to minimize the total energy expended in returning to its nesting area?
 - Let W and L denote the energy (in joules) per kilometer flown over water and land, respectively. What would a large value of the ratio W/L mean in terms of the bird's flight? What would a small value mean?

Determine the ratio W/L corresponding to the minimum expenditure of energy.

- What should the value of W/L be in order for the bird to fly directly to its nesting area D? What should the value of W/L be for the bird to fly to B and then along the shoreline to D?
- If the ornithologists observe that birds of a certain species reach the shore at a point 4 km from B, how many times more energy does it take a bird to fly over water than over land?



- 30. Crows and whelks** Crows on the west coast of Canada feed on whelks by carrying them to heights of about 5 m and dropping them onto rocks (several times if necessary) to break open their shells. Two of the questions raised by the author of a study of this phenomenon were “Do crows drop whelks from the best height for breaking?” and “How energetically profitable is dropping of whelks?” The author constructed poles and dropped whelks from various heights. A model based on the study's data for the number of times a whelk needs to be dropped from a height h to be broken is

$$n(h) = \frac{h + 14.8}{h - 1.2}$$

where h is measured in meters. The energy expended by a crow in this activity is proportional to the height h and to the number $n(h)$:

$$E = khn(h) = \frac{kh(h + 14.8)}{h - 1.2}$$

- What value of h minimizes the energy expended by the crows?
- How does your answer to part (a) compare with the observed average dropping height of 5.3 m that is actually used by crows? Does the model support the existence of an optimal foraging strategy?

Source: Adapted from R. Zach, “Decision-making and Optimal Foraging in Northwestern Crows,” *Behaviour* 68 (1979): 106–17.

PROJECT Flapping and Gliding



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FIGURE 1



Small birds like finches alternate between flapping their wings and keeping them folded while gliding (see Figure 1). In this project we analyze this phenomenon and try to determine how frequently a bird should flap its wings. Some of the principles are the same as for fixed-wing aircraft and so we begin by considering how required power and energy depend on the speed of airplanes.¹

1. The power needed to propel an airplane forward at velocity v is

$$P = Av^3 + \frac{BL^2}{v}$$

where A and B are positive constants specific to the particular aircraft and L is the lift, the upward force supporting the weight of the plane. Find the speed that minimizes the required power.

2. The speed found in Problem 1 minimizes power but a faster speed might use less fuel. The energy needed to propel the airplane a unit distance is $E = P/v$. At what speed is energy minimized?
3. How much faster is the speed for minimum energy than the speed for minimum power?
4. When applying the equation of Problem 1 to bird flight, we split the term Av^3 into two parts: $A_b v^3$ for the bird's body and $A_w v^3$ for its wings. Let x be the fraction of flying time spent in flapping mode. If m is the bird's mass and all the lift occurs during flapping, then the lift is mg/x , where g is the acceleration due to gravity, and so the power needed during flapping is

$$P_{\text{flap}} = (A_b + A_w)v^3 + \frac{B(mg/x)^2}{v}$$

The power while wings are folded is $P_{\text{fold}} = A_b v^3$. Show that the average power over an entire flight cycle is

$$\bar{P} = xP_{\text{flap}} + (1 - x)P_{\text{fold}} = A_b v^3 + xA_w v^3 + \frac{Bm^2 g^2}{xv}$$

5. For what value of x is the average power a minimum? What can you conclude if the bird flies slowly? What can you conclude if the bird flies increasingly faster?
6. The average energy over a cycle is $\bar{E} = \bar{P}/v$. What value of x minimizes $\bar{E}$?
7. Compare your answers to Problems 5 and 6. What do you notice?

1. Adapted from R. Alexander, *Optima for Animals* (Princeton, NJ: Princeton University Press, 1996).

PROJECT The Tragedy of the Commons: An Introduction to Game Theory



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In Example 4.4.5 we explored sustainable fish harvesting. We assumed that a single company is exploiting the resource and found that the steady-state population size in the presence of harvesting satisfied the equation

$$rN\left(1 - \frac{N}{K}\right) = hN$$

where N is the population size, r and K are positive constants, and h is the fishing effort. In reality, fish stocks are part of the “Commons,” meaning that no single person has exclusive rights to them. Suppose, for example, that a second company begins to exploit the same population. Then there are two fishing efforts, h_1 and h_2 , one for each company. Once the population size has stabilized, the equation

$$rN\left(1 - \frac{N}{K}\right) = h_1N + h_2N$$

must hold, where h_1N and h_2N are the total harvests for companies 1 and 2, respectively. Suppose you run company 1 and before company 2 arrives you are using the optimal h calculated in Example 4.4.5, that is, $h_1 = \frac{1}{2}r$.

1. When company 2 arrives, it needs to decide upon a fishing effort h_2 . What value of h_2 maximizes its harvest once the population has reached a steady state, assuming that you continue using h_1 ?
2. Once your competitor is using their rate obtained in Problem 1, your harvesting rate will no longer be optimal for you. What is your new optimal rate h_1^* , given that your competitor continues to use the rate found in Problem 1?
3. More generally, determine your optimal fishing effort as a function of the rate your competitor uses and your competitor’s optimal fishing effort as a function of the rate you use. These are referred to as the “best response” fishing efforts.
4. The harvesting problem can be viewed as a game played between the two companies, where the payoff to each depends on both of their choices of fishing effort. An area of mathematics called **game theory** has been developed to analyze such problems. A key concept in game theory is that of a **Nash equilibrium**, which is a pair of values h_1^* and h_2^* that simultaneously satisfy both best response functions. At a Nash equilibrium each party is doing the best that it can, given the choice of its competitor. What is the Nash equilibrium pair of fishing efforts?
5. What is the total population size at the Nash equilibrium, and what are the total harvests of you and your competitor?
6. Demonstrate that both you and your competitor could have a higher total harvest than that attained at the Nash equilibrium if you could agree to cooperate and to split the harvest that you were obtaining before your competitor showed up.
7. Problem 6 shows that both you and your competitor are worse off at the Nash equilibrium than you would be if you agreed to cooperate. Show that, in terms of population size, the fish population is also worse off. This is the “Tragedy of the Commons.”

Nash

John F. Nash, Jr. (1928–) is an American mathematician best known for his work in game theory. He developed the idea now known as a Nash equilibrium in his 28-page doctoral thesis in 1950. In 1994 he was awarded the Nobel Prize in Economics for this work. Nash also made several other foundational contributions to advanced mathematics, despite suffering from schizophrenia. His extraordinary life is chronicled in the book *A Beautiful Mind* by Sylvia Nasar and in a Hollywood movie of the same name.

4.5 Recursions: Equilibria and Stability

In Section 1.6 we looked at recursive sequences, which we also called difference equations or discrete-time models. These are defined by a recursion of the form

$$a_{n+1} = f(a_n) \quad \text{or} \quad x_{t+1} = f(x_t) \quad \text{or} \quad N_{t+1} = f(N_t)$$

where f is the updating function, N_t is the number of individuals in a population at time t , and N_{t+1} is the population one unit of time into the future. Then in Section 2.1 we investigated the long-term behavior of such recursions. In particular, we saw that some recursive sequences approach a limiting value as t becomes large:

$$\lim_{t \rightarrow \infty} x_t = L$$

Here we assume that the updating function f that defines the recursion is a differentiable function and learn that the values of its derivative play a role in determining the limiting behavior of the sequence.

Equilibria

(1) Definition An **equilibrium** of a recursive sequence $x_{t+1} = f(x_t)$ is a number $\hat{x}$ that is left unchanged by the function f , that is,

$$f(\hat{x}) = \hat{x}$$

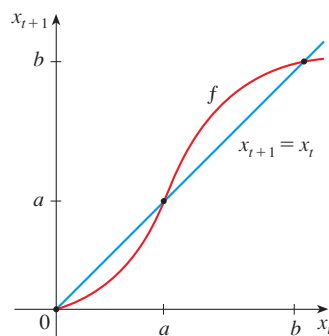


FIGURE 1

The recursion $x_{t+1} = f(x_t)$ has three equilibria

It's helpful to think of an equilibrium as a point on a number line. An equilibrium is sometimes called a **fixed point** because f leaves the point $\hat{x}$ fixed. Notice that if $\hat{x}$ is an equilibrium and if, for instance, $x_6 = \hat{x}$, then

$$x_7 = f(x_6) = f(\hat{x}) = \hat{x}$$

and, similarly, all of the following terms in the sequence are also equal to $\hat{x}$.

To find the equilibria algebraically, we solve the equation $f(x) = x$, if possible. To locate them geometrically we graph the curves $y = f(x)$ and $y = x$ (the diagonal line) and see where they intersect. Because the recursion is $x_{t+1} = f(x_t)$, when we graph f we label the horizontal and vertical axes x_t and x_{t+1} , as in Figure 1. For that particular recursion we see that there are three points of intersection and therefore three equilibria, 0, a , and b .

(2) Definition An equilibrium is called **stable** if solutions that begin close to the equilibrium approach that equilibrium. It is called **unstable** if solutions that start close to the equilibrium move away from it.

So when we say that $\hat{x}$ is a stable equilibrium of the recursion $x_{t+1} = f(x_t)$ we mean that if x_t is a solution of the recursion and x_0 is sufficiently close to $\hat{x}$, then $x_t \rightarrow \hat{x}$ as $t \rightarrow \infty$.